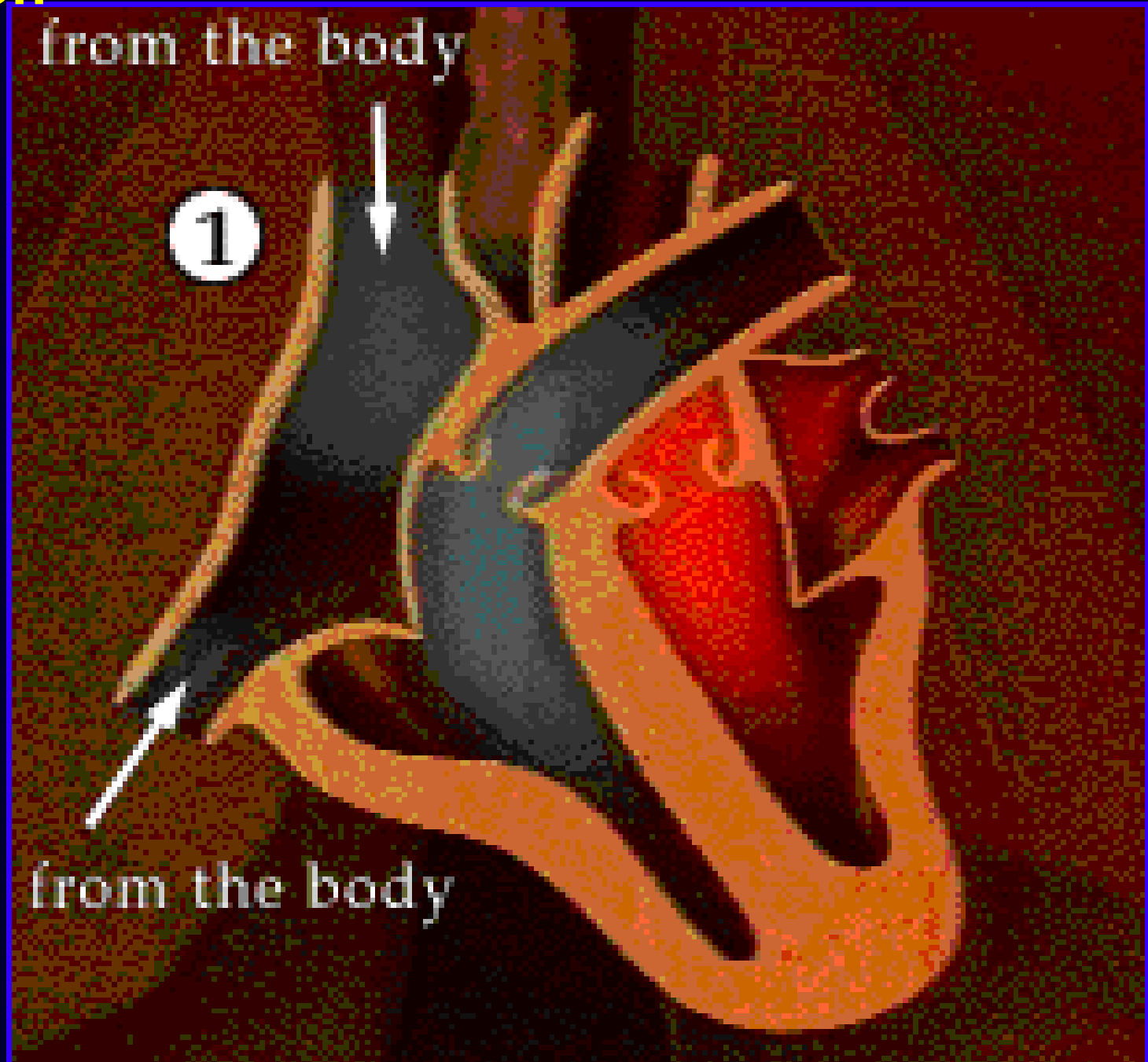




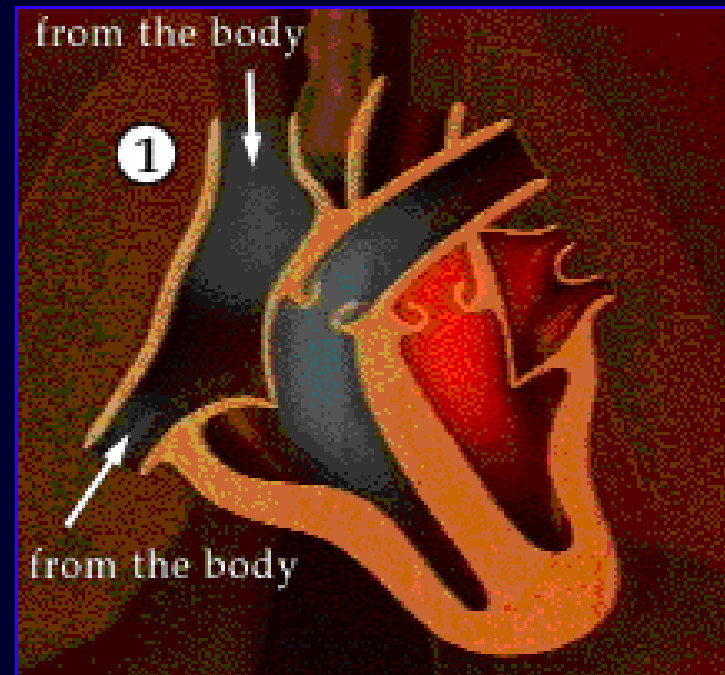
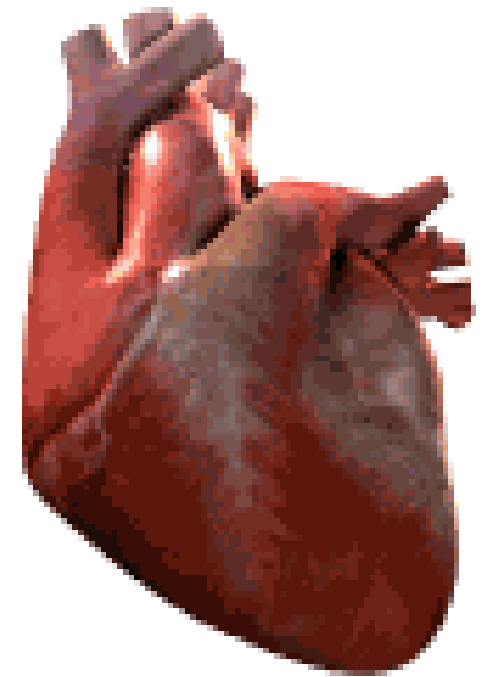
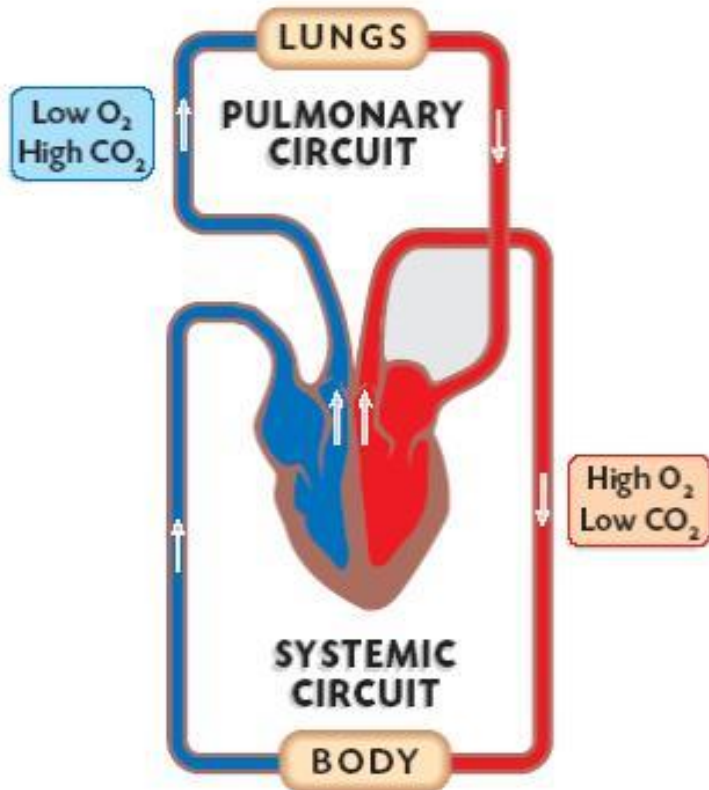
CARDIOVASCULAR SYSTEM

Dr. Mohamed Zidan
mohamedzidan@alexu.edu.eg

Circulation



Circulation



Cardiovascular system

Heart

Wall of the heart

Cardiac skeleton

Conducting system

Valves of the heart

Chordae tendineae

Hormones of the heart

**Sensory
receptors**

Carotid body

Carotid sinus

**Blood
vessels**

Arteries

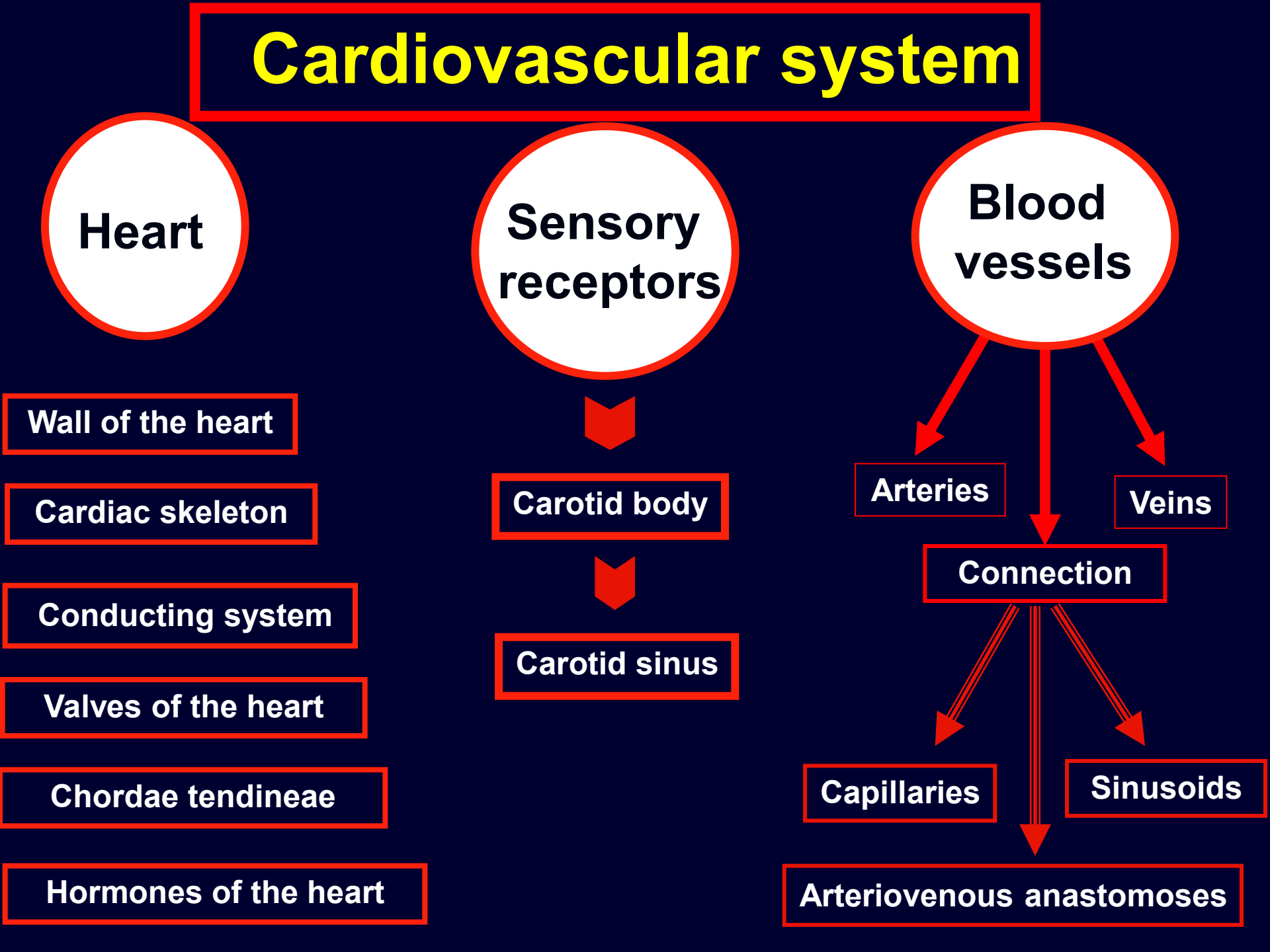
Veins

Connection

Capillaries

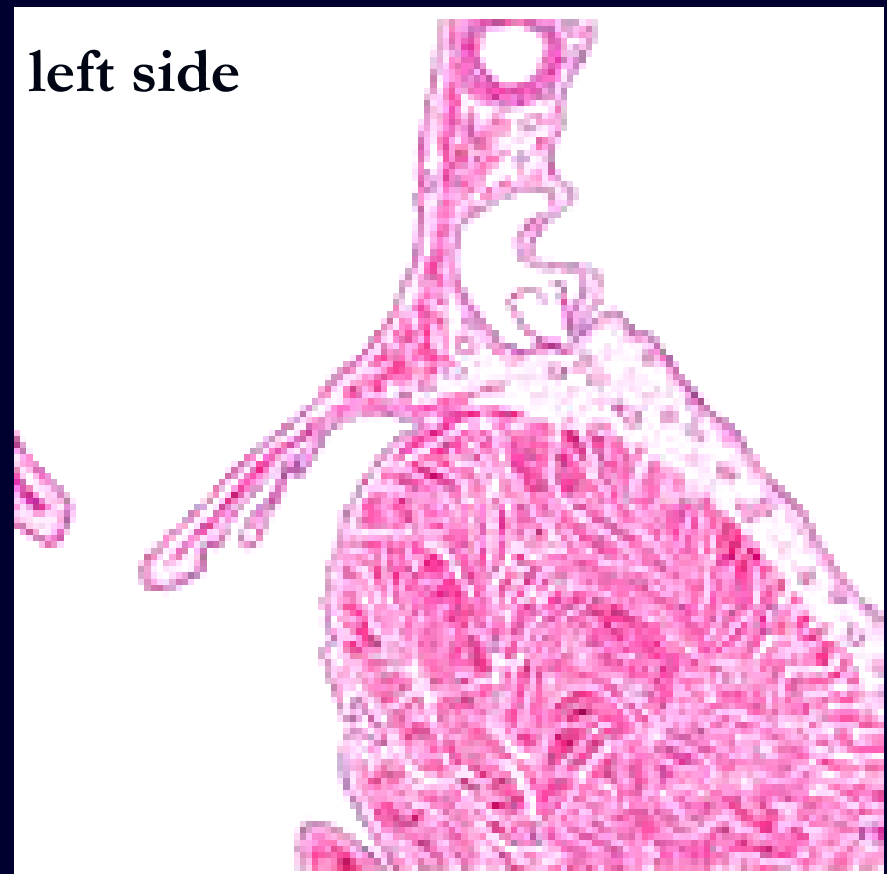
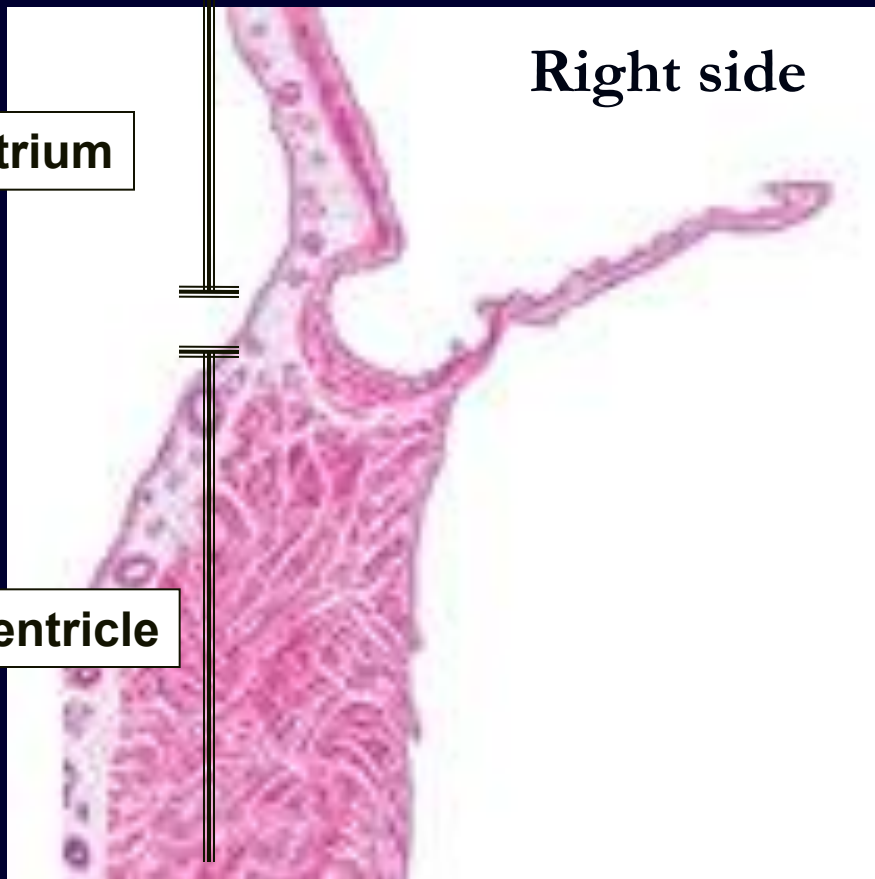
Sinusoids

Arteriovenous anastomoses

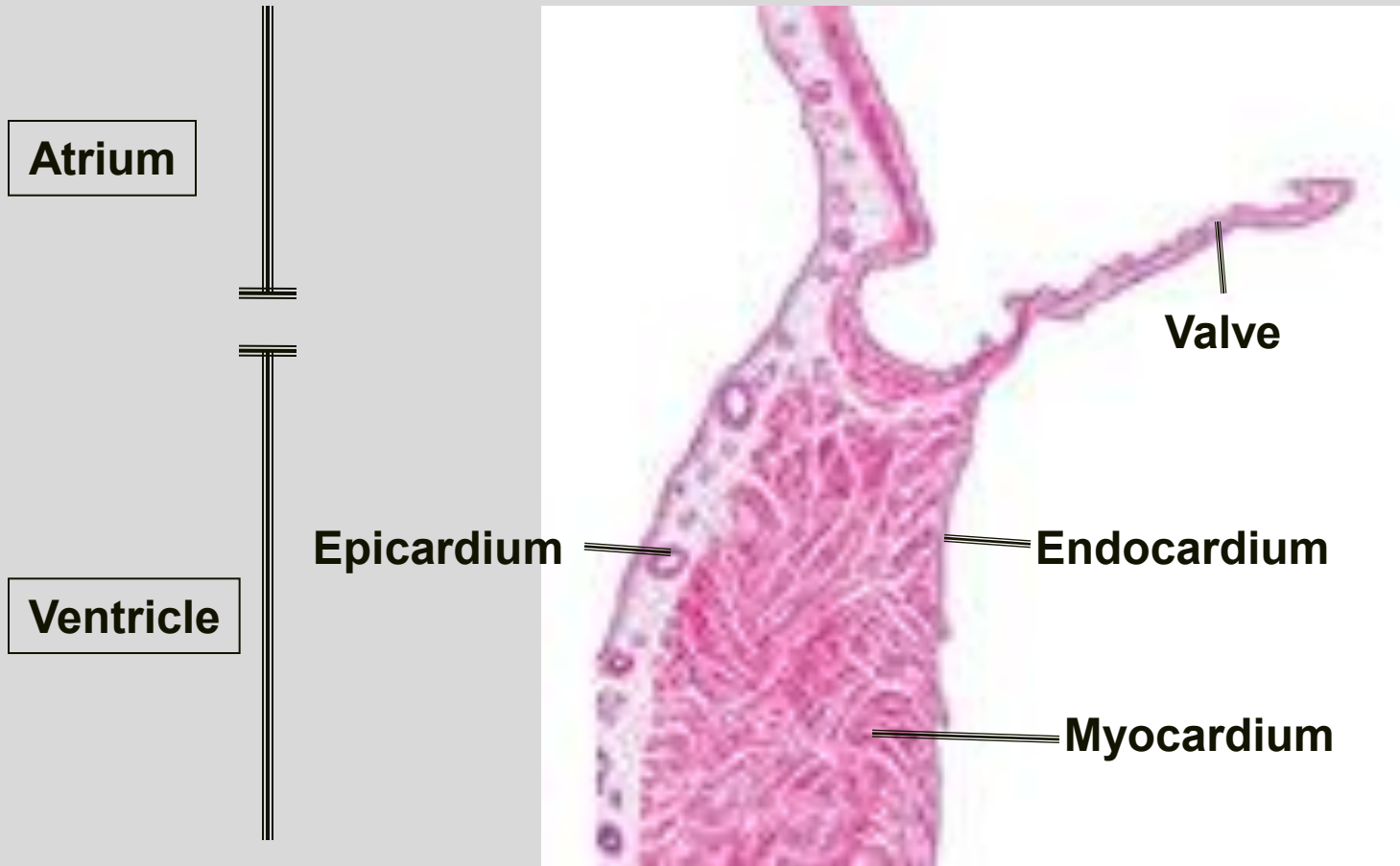


Heart wall

- The wall of the atria is thinner than that of the ventricles.
- The wall of right atrium is thinner than the wall of left atrium.



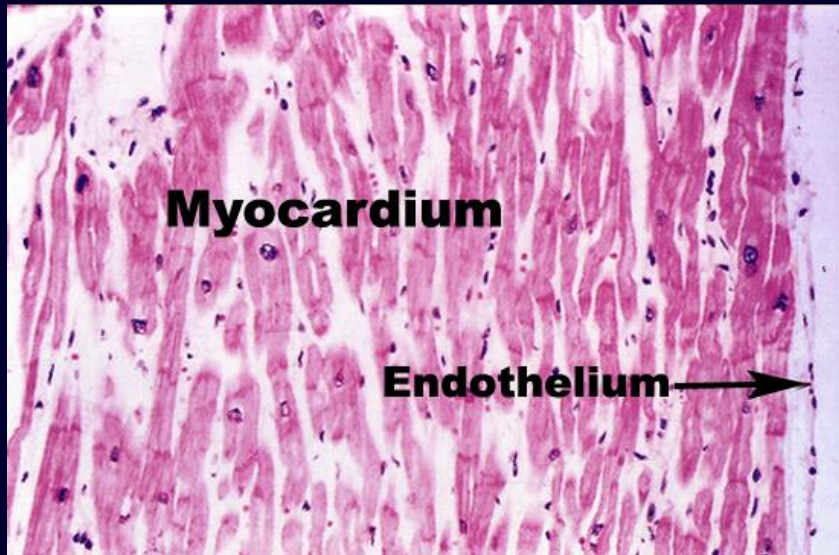
Heart wall



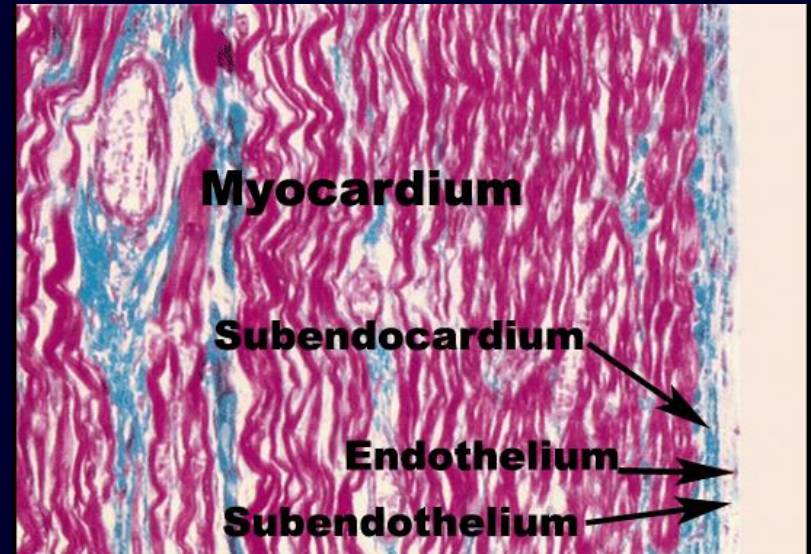
1. Endocardium

- The **endocardium** lines all the internal surfaces of the heart. It consists of three layers:
- 1. **The endothelium** of simple squamous epithelium.
- 2. **The subendothelium** of thin delicate connective tissue that contains **numerous elastic fibers** and **few smooth muscles**.
- 3. **The subendocardium** is the site of connection between the endocardium and the myocardium. It consists of loose connective tissue containing **blood vessels**, **nerves** and **branches of the impulse conducting system** of the heart.

Endocardium



H&E stain

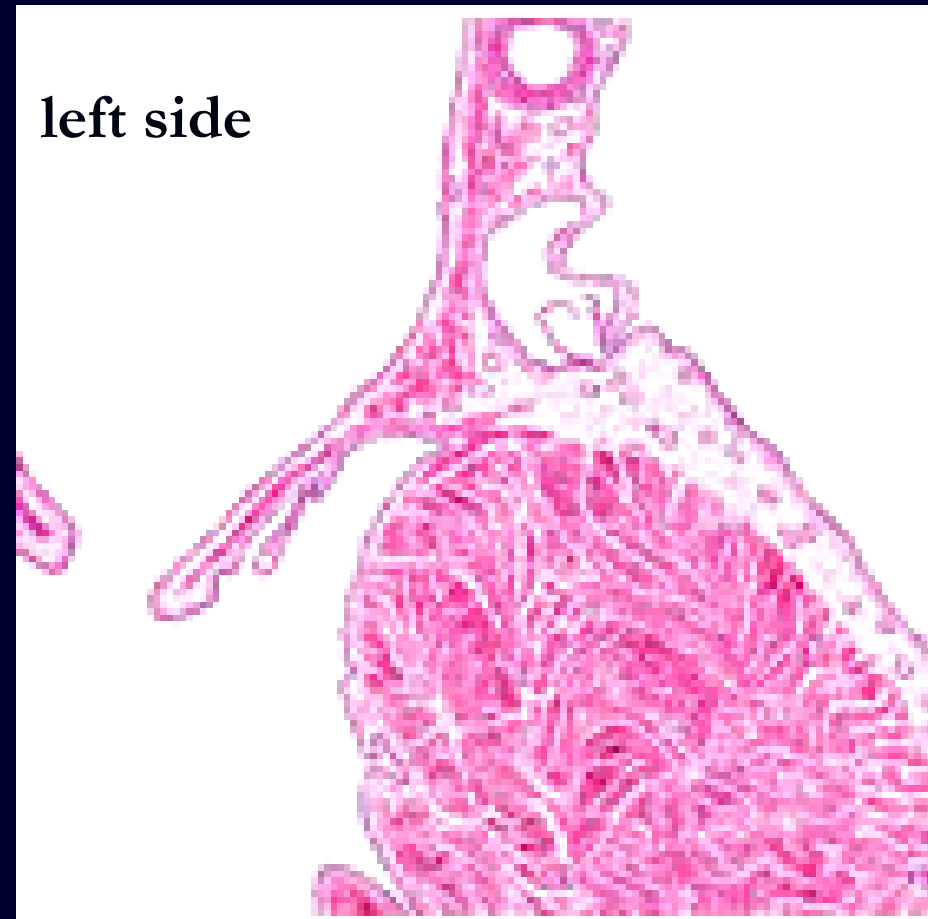
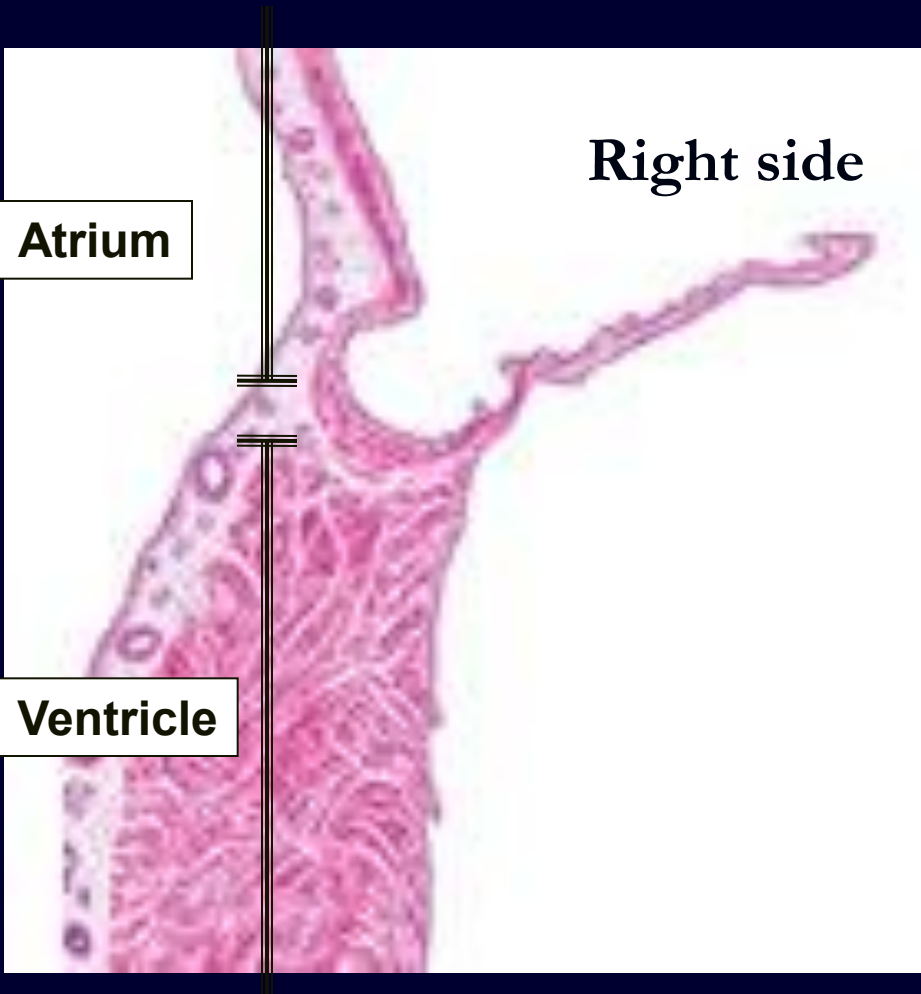


Masson's Trichrome stain

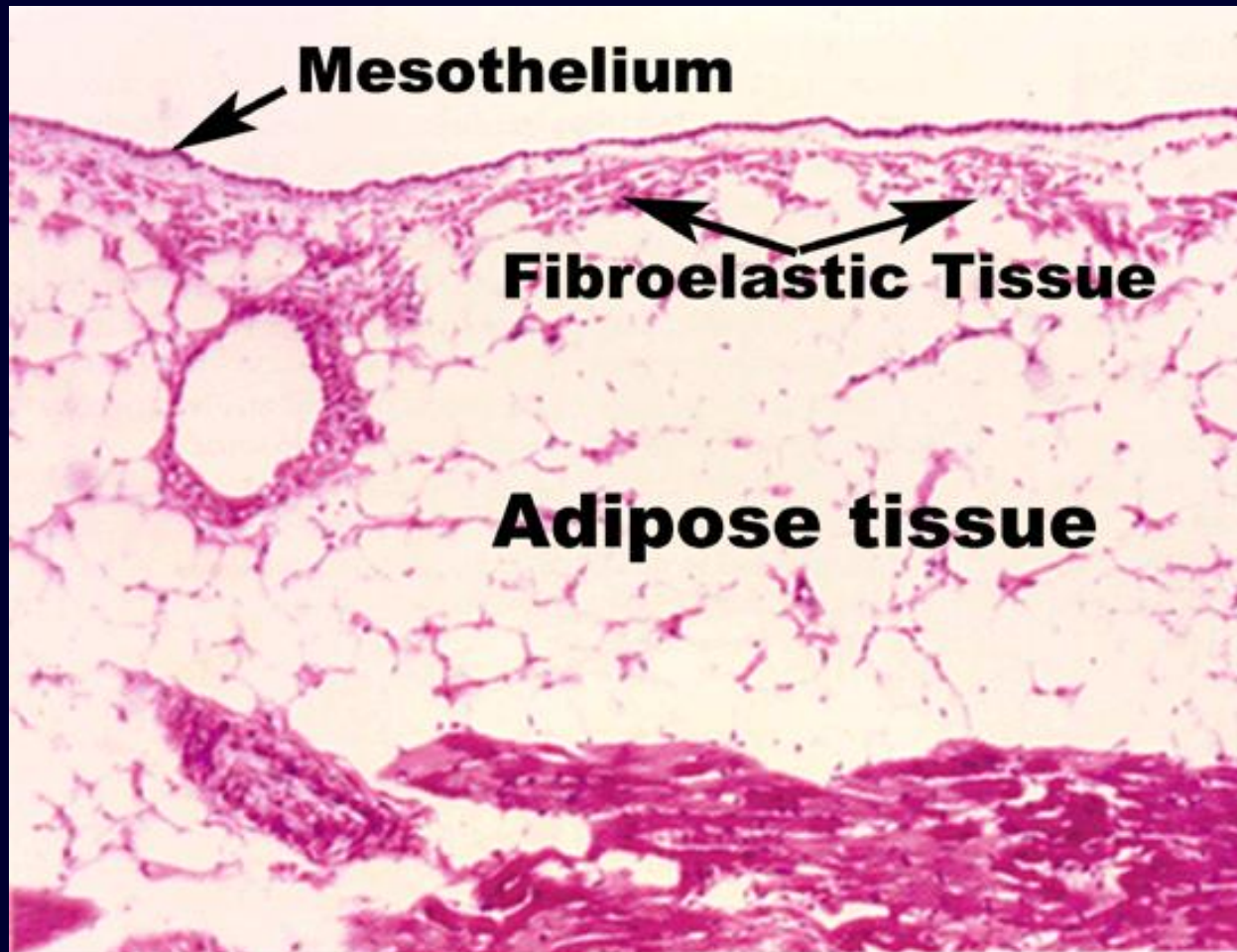
2. Myocardium

- The myocardium forms the main layer of the heart wall and is composed of **cardiac muscles** attached to the **fibrous skeleton** of the heart.
- In the atria, the myocardium is thinner than in the ventricles. The thickness of the myocardium depends on the pressure in each chamber. So that the **thickest myocardium is found in the left ventricle** .
- The cardiac muscles of **the atria** and **ventricles** are completely separated by **fibrous skeleton** of the heart, the only connection being the conducting system of the heart.

2. Myocardium



Epicardium



3. Epicardium

- The epicardium is the outermost covering of the heart.
- It forms the **visceral layer of the pericardium** and consists of :
 1. **Flattened mesothelial cells** resting on
 2. A layer of connective tissue which **contains nerves and the coronary arteries and veins.**
- Adipose tissue is sometimes found in large amounts around the coronary blood vessels.

Cardiac skeleton

```
graph TD; A[Cardiac skeleton] --> B[Anuli fibrousi (fibrous rings)]; A --> C[Trigonum fibrosum (triangular mass)]; A --> D[Inter ventricular septum];
```

Anuli fibrousi (fibrous rings)

- The atrioventricular openings
- The openings of the aorta and pulmonary artery.

Collagen bundles and few elastic fibers.

Trigonum fibrosum (triangular mass)

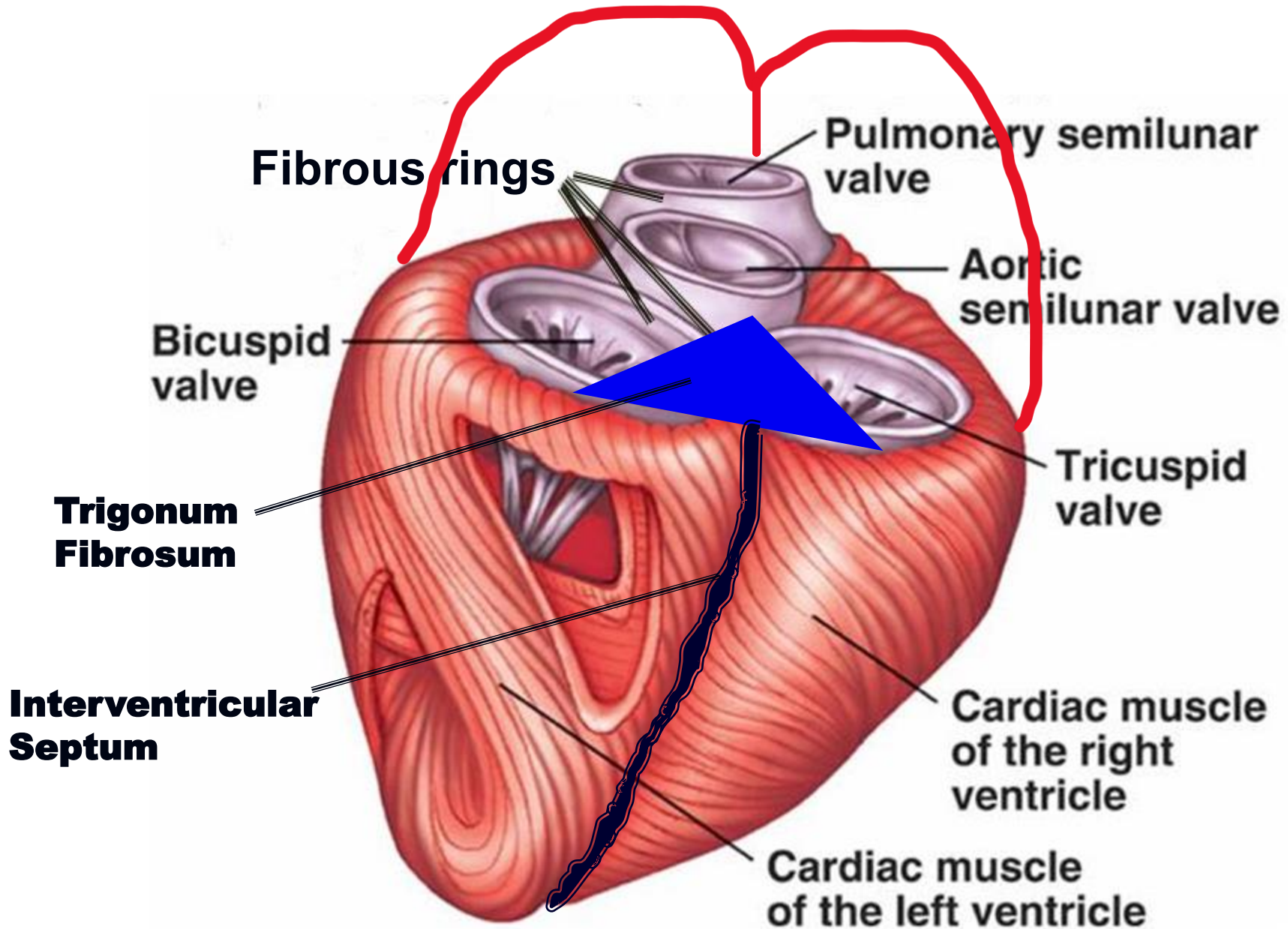
Between the atrioventricular openings and the base of the Aorta.

- Dense irregular C.T.: (pig & cat)
- Fibrocartilage: (dog)
- Hyaline cartilage: (Horse)
- bone: (large ruminants)

Inter ventricular septum

Collagen bundles.

Cardiac skeleton



Os Cordis



Impulse conducting system

Composed of a special type of cardiac muscles which are specialized for either initiation or conduction of impulses for contraction of the heart and consists of:

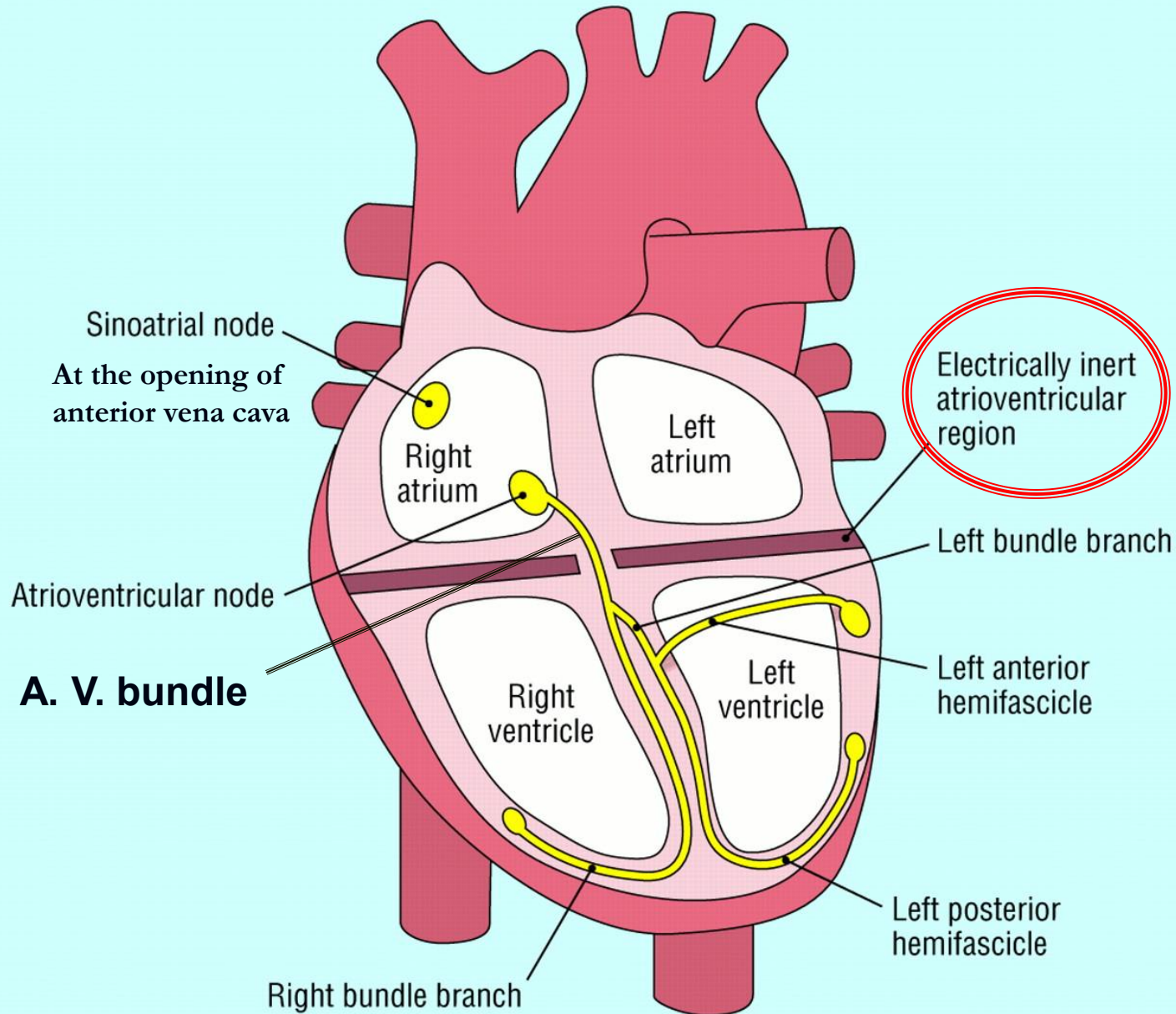
**The sinoatrial node
(SA node)**

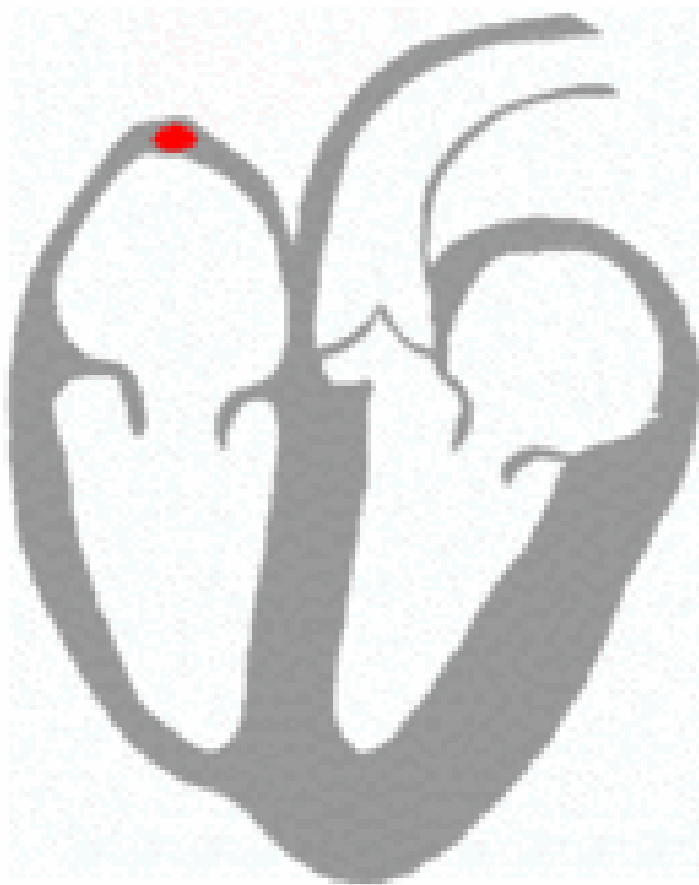
**The atrioventricular node
(AV node)**

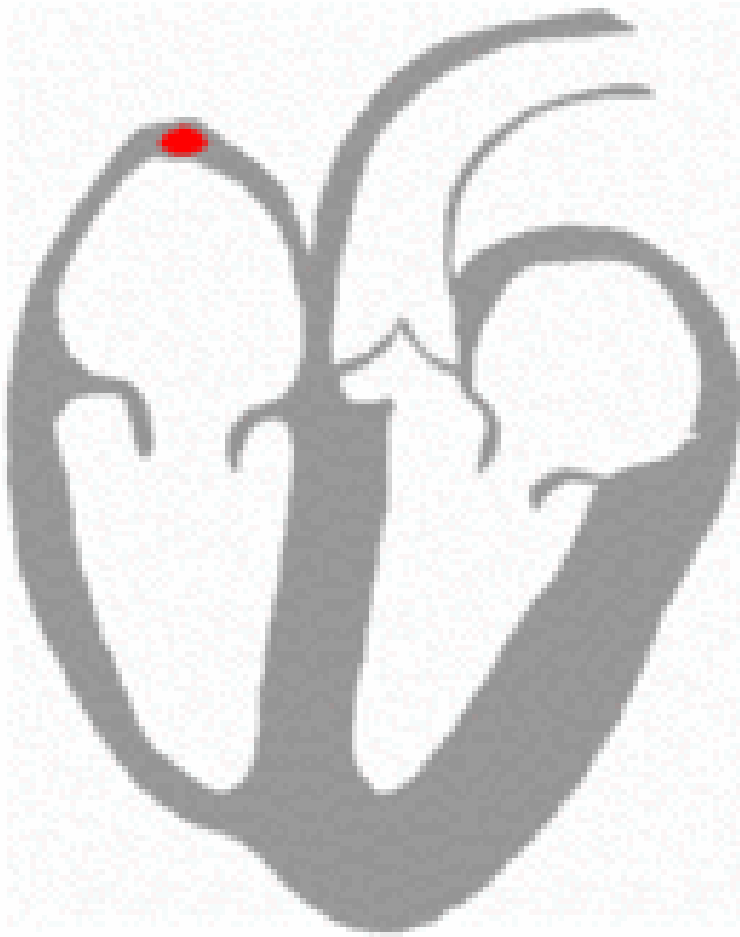
**The atrioventricular bundle
(AV bundle)**

Purkinje fibers

Impulse conducting system

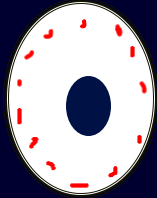






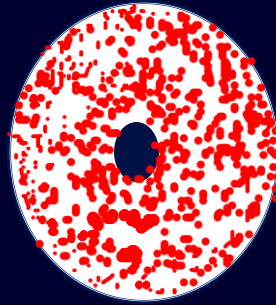
Principle of ECG formation. Note that the red lines represent the depolarization wave.

Impulse conducting system Cells

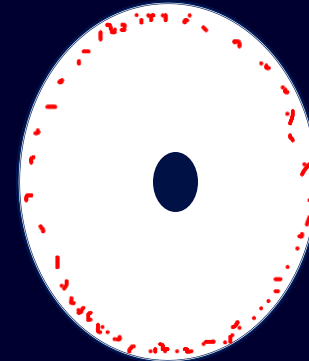


SA. & AV. nodes

**Smaller
Paler
Few myofibrils
Rich in glycogen**



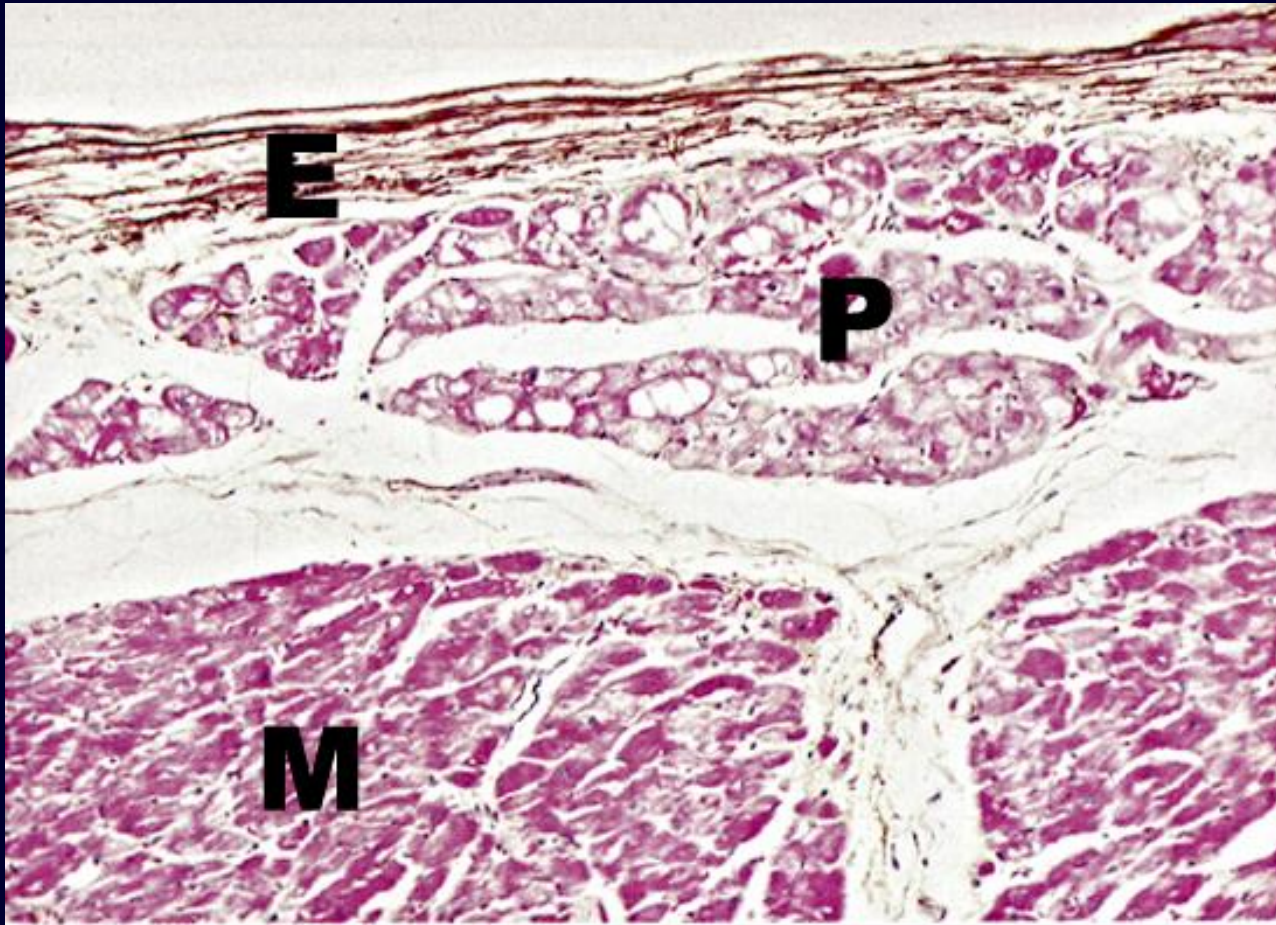
Cardiac Muscles



Purkinje Fibers

**Larger
Paler
Few myofibrils
Rich in glycogen**

Purkinje Fibers



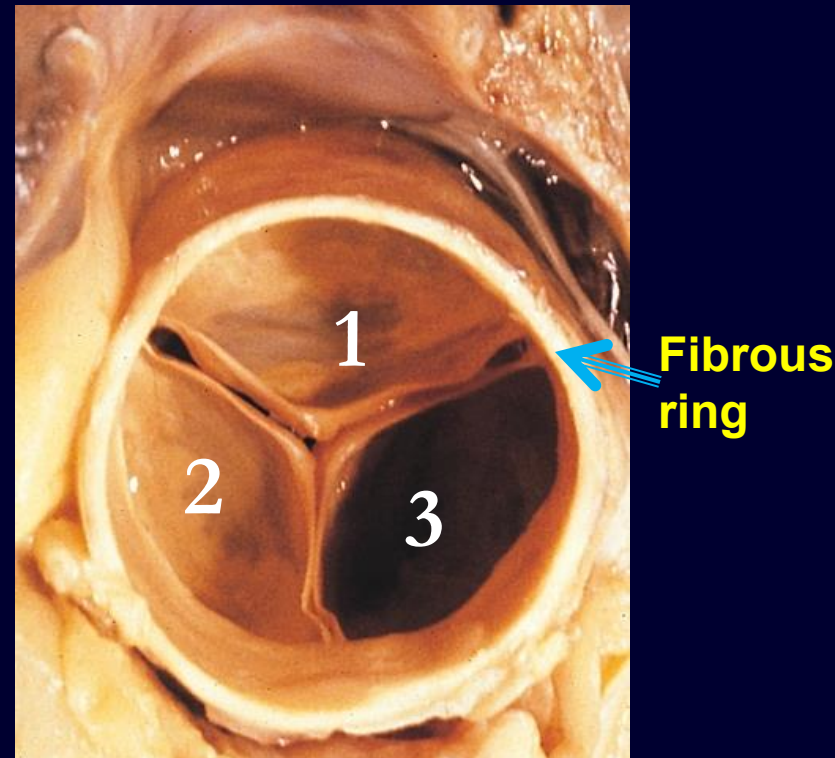
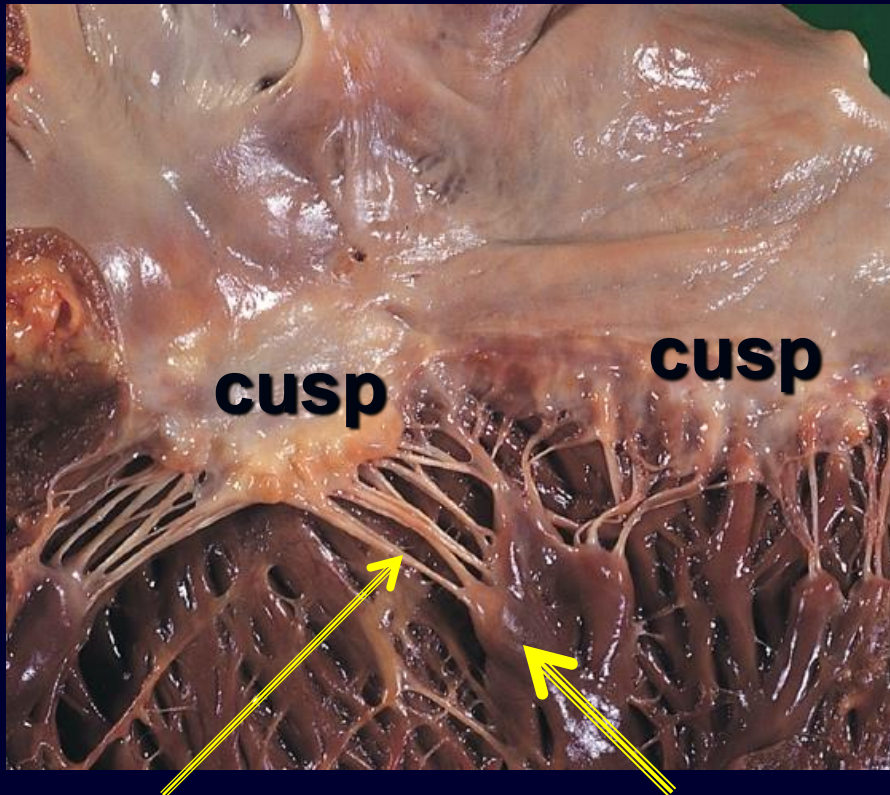
Valves of the heart

The heart contains 4 valves:

- **Tricuspid valve** : guards the right atrioventricular orifice. It has 3 cusps which attached by their bases to the fibrous ring of the heart skeleton.
- **Mitral valve** : guards the left atrioventricular orifice. It consists of 2 cusps which attached by their bases to the fibrous ring.
- **Pulmonary valve**: guards the pulmonary orifice. It consists of 3 semilunar cusps.
- **Aortic valve**: guards the aortic orifice. It consists of 3 semilunar cusps.

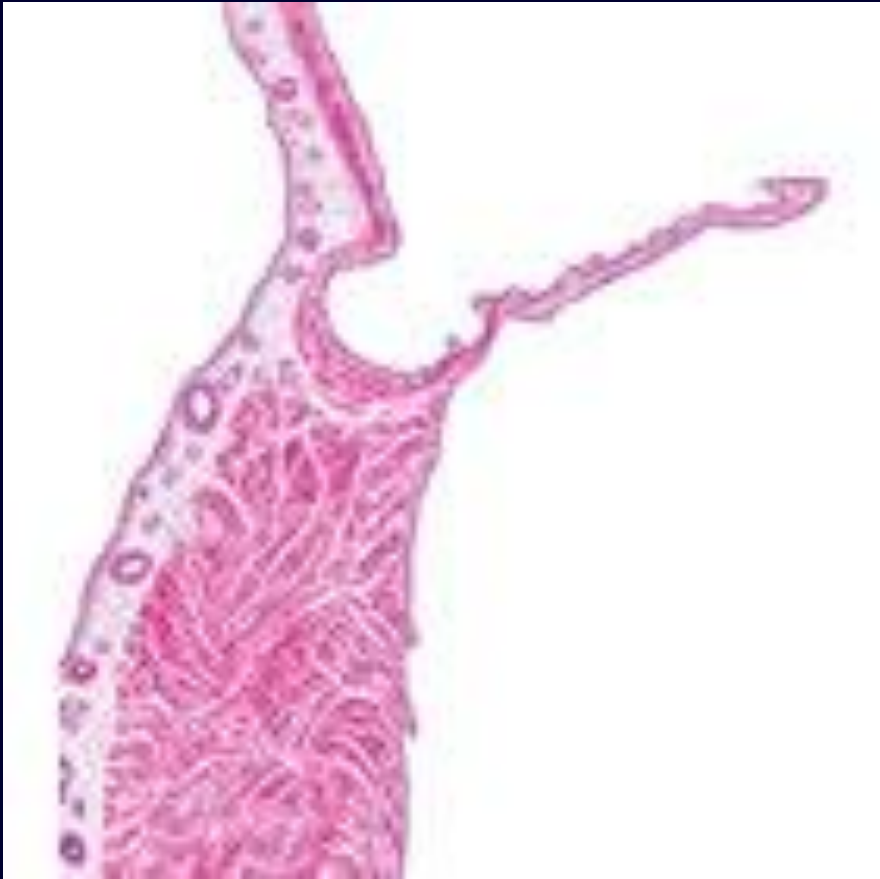
Valves

Each valves consists of a number of cusps each cusp composed of a central core of dense irregular connective tissue covered by endocardium.

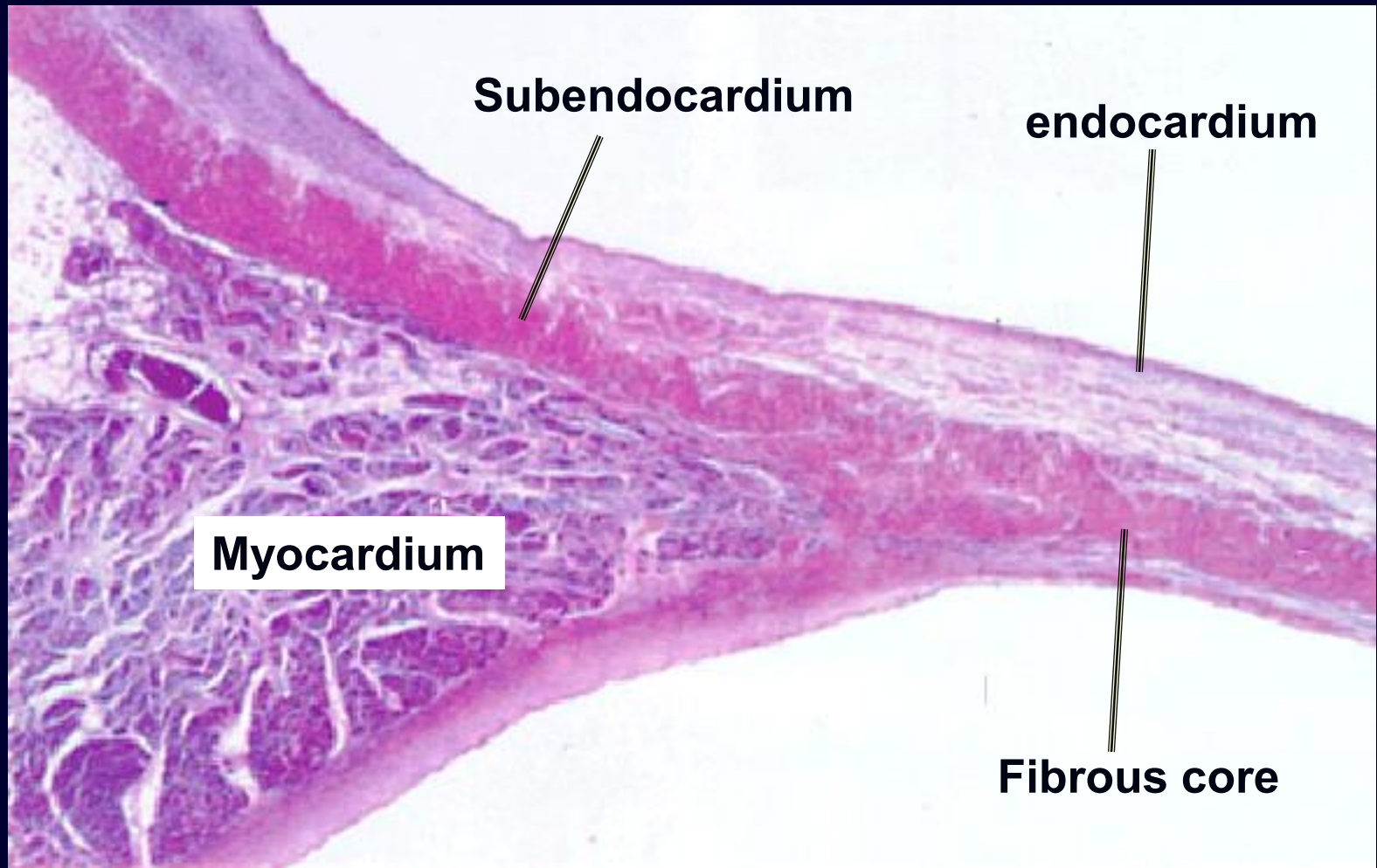


Tricuspid valve

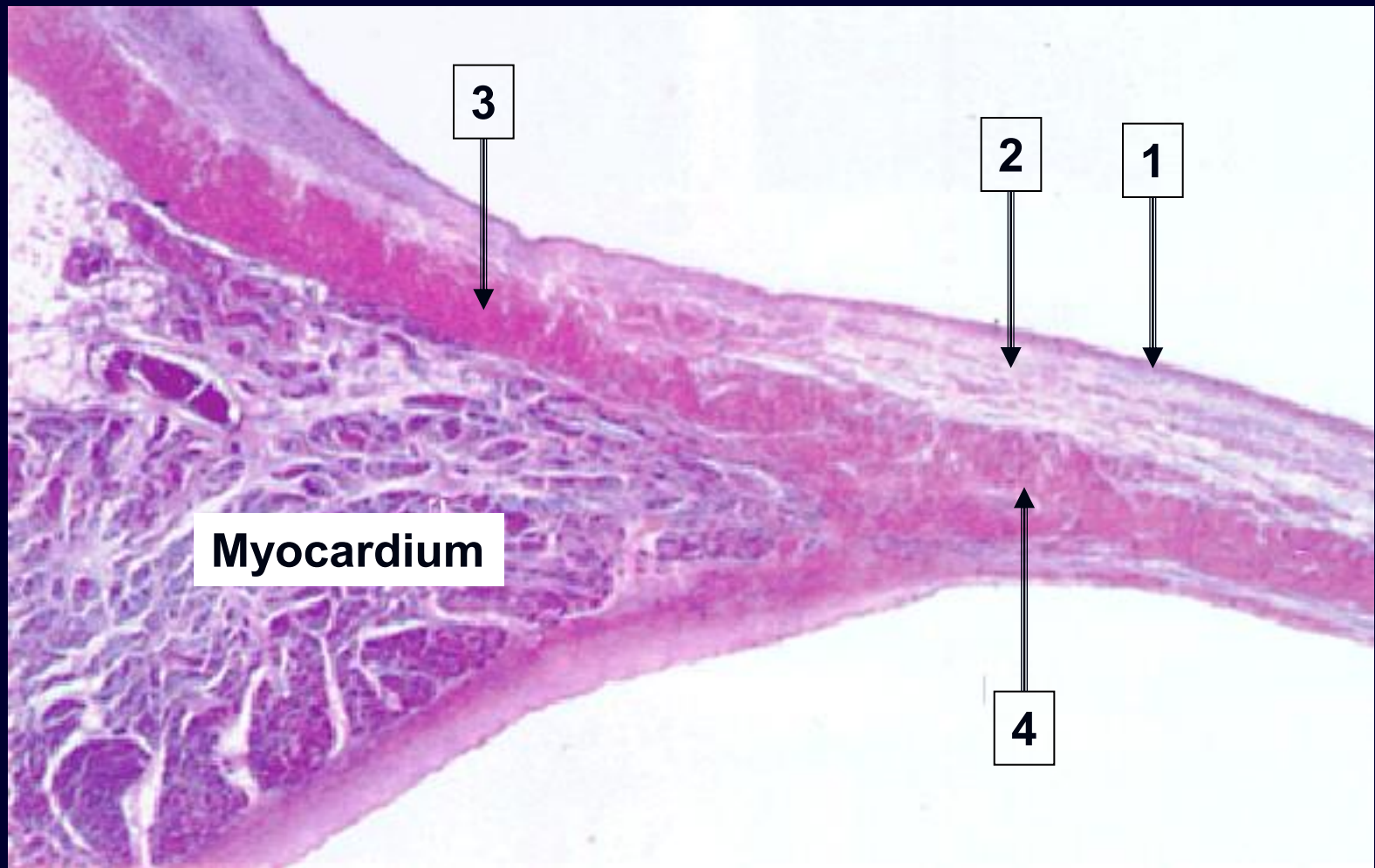
Heart



Valve



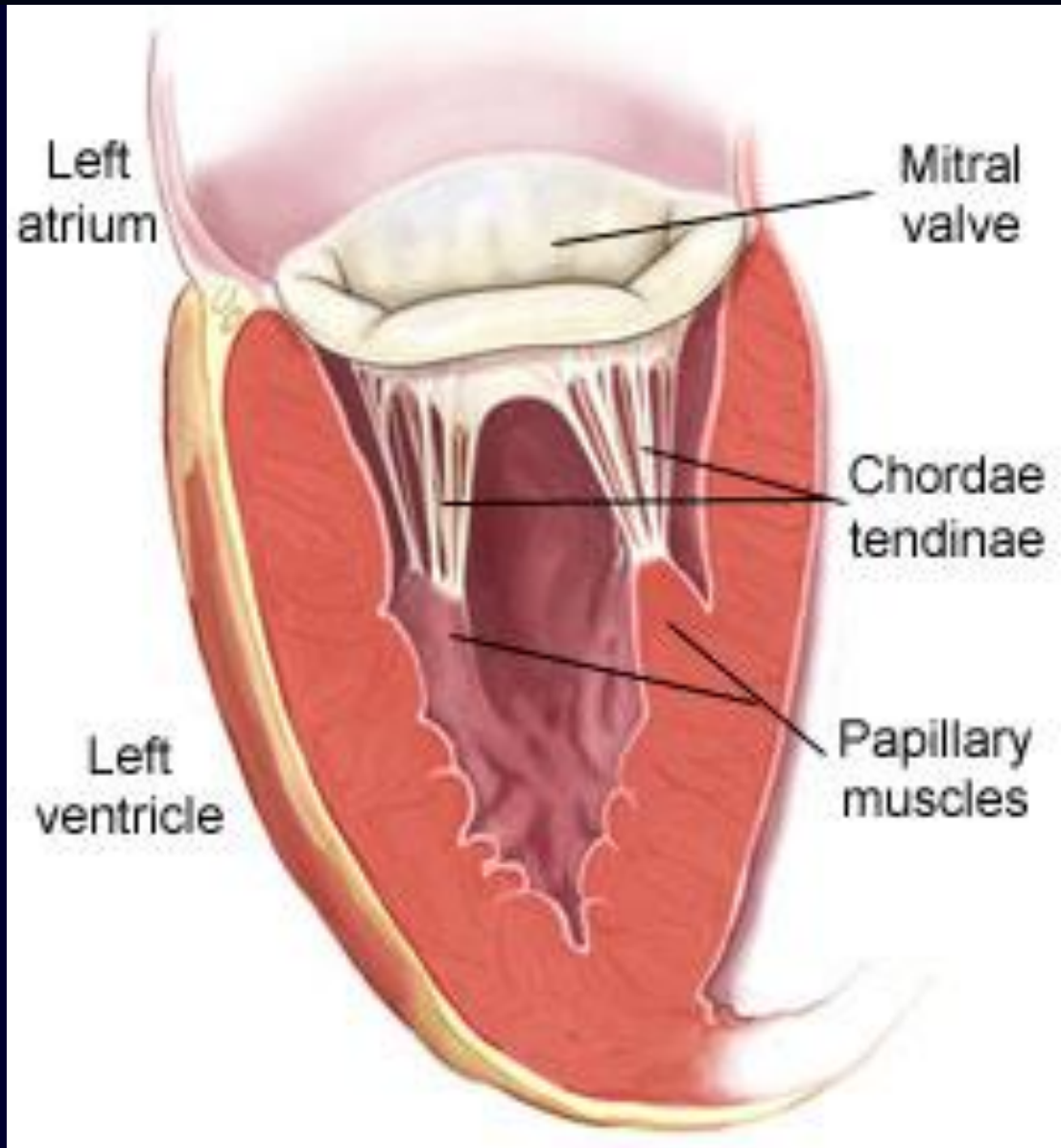
Valve



Chordae tendineae

- The margins and ventricular surfaces of the cusps of both the tricuspid and mitral valves are connected to small muscular projections on the ventricular walls (**papillary muscles**) by a number of tendinous cords; the chordae tendineae.
- These cords are composed of dense fibrous tissue and covered with endocardium.

Chordae tendineae



Hormones of the heart

The right atrial muscle cells produce a polypeptide hormone called atrial natriuretic factor (ANF) which is stored in membrane bound granules in atrial cardiocytes in the perinuclear zone in association with prominent Golgi complex and numerous rough endoplasmic reticulum.

ANF

Regulate blood pressure

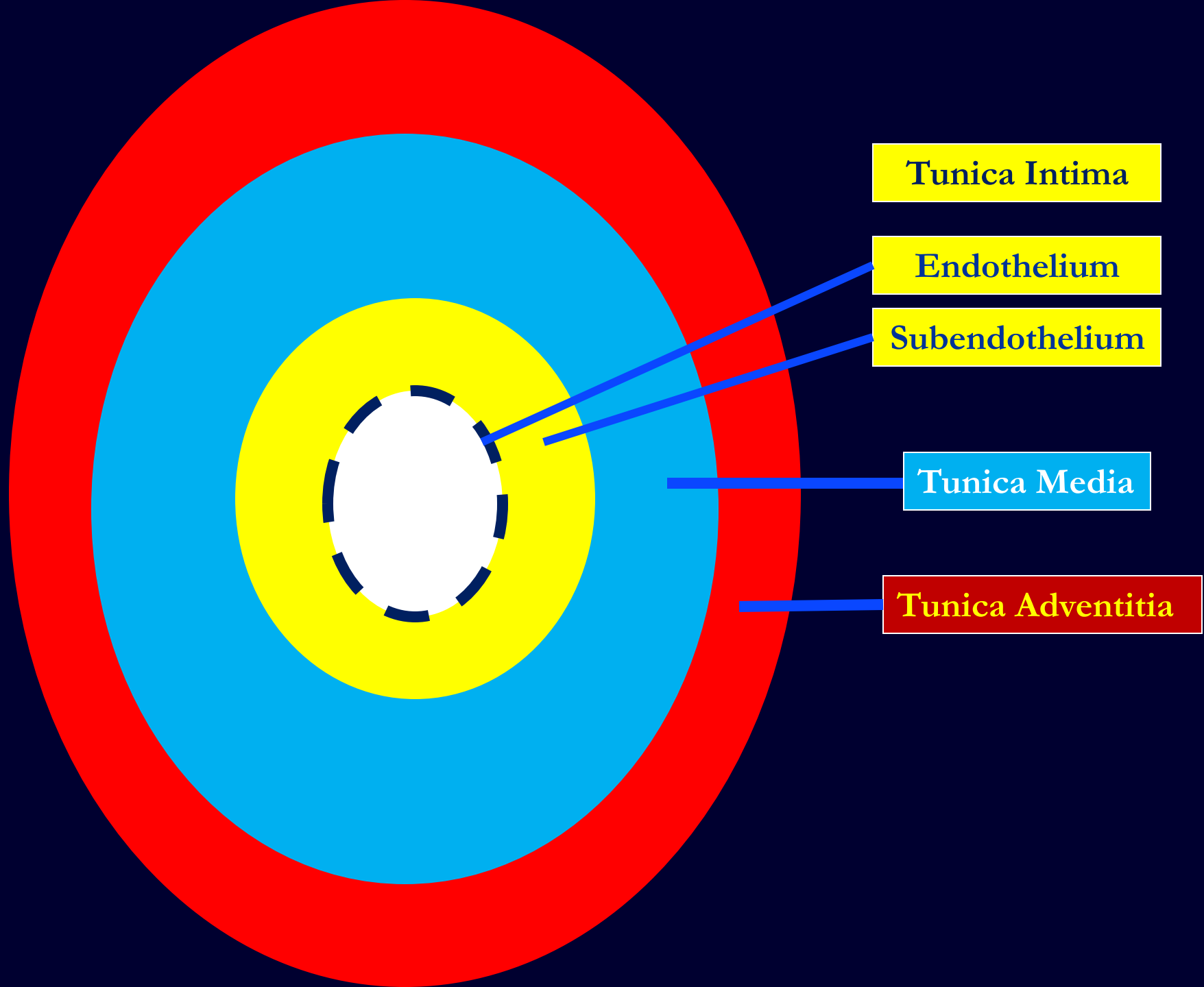
Cardionatriin

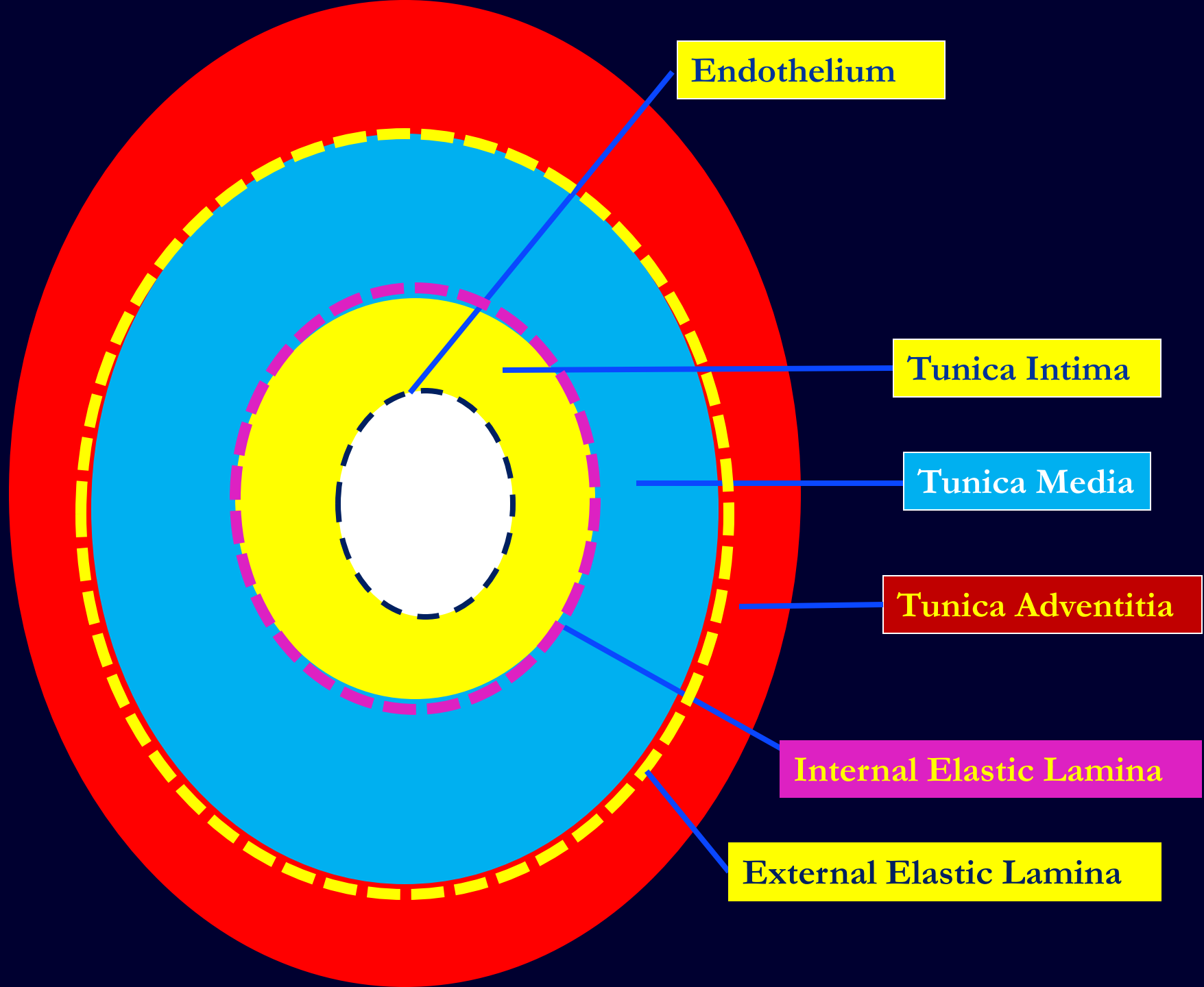
Potent diuretic and natriuretic effects.

Cardiodilatin

**Vascular smooth muscle relaxation
(Vasodilatation)**

Blood Vessels





Structure of blood vessels

Tunica intima (L. internal coat)

Endothelium

Subendothelium (Thin layer of loose C.T.)

Internal elastic lamina

Tunica media (middle coat)

Smooth muscles

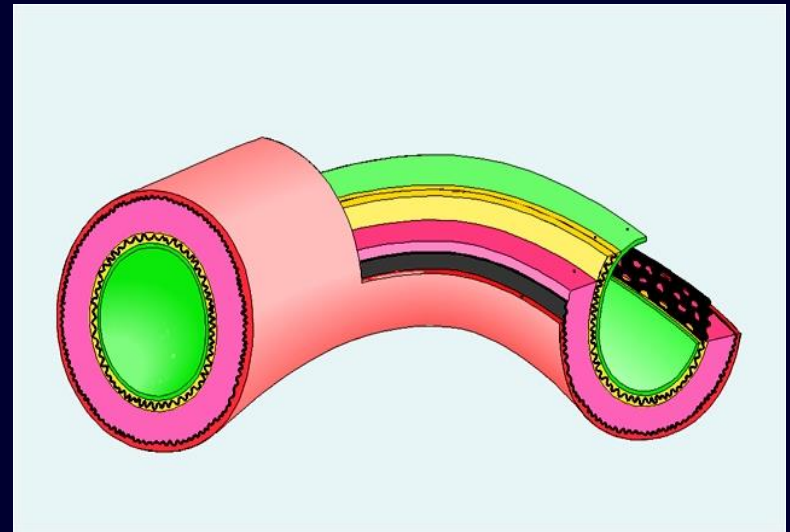
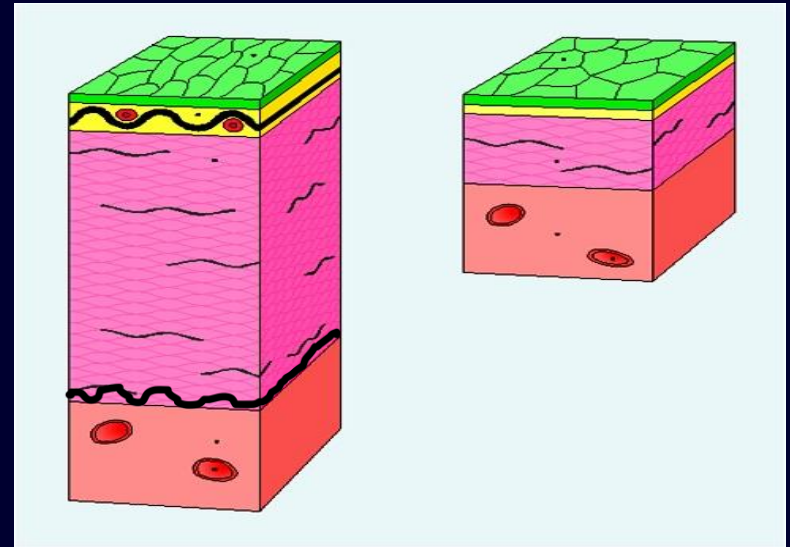
Elastic fibers

Collagen fibers

External elastic lamina

Tunica adventitia (outer coat)

Loose C.T.



Arteries

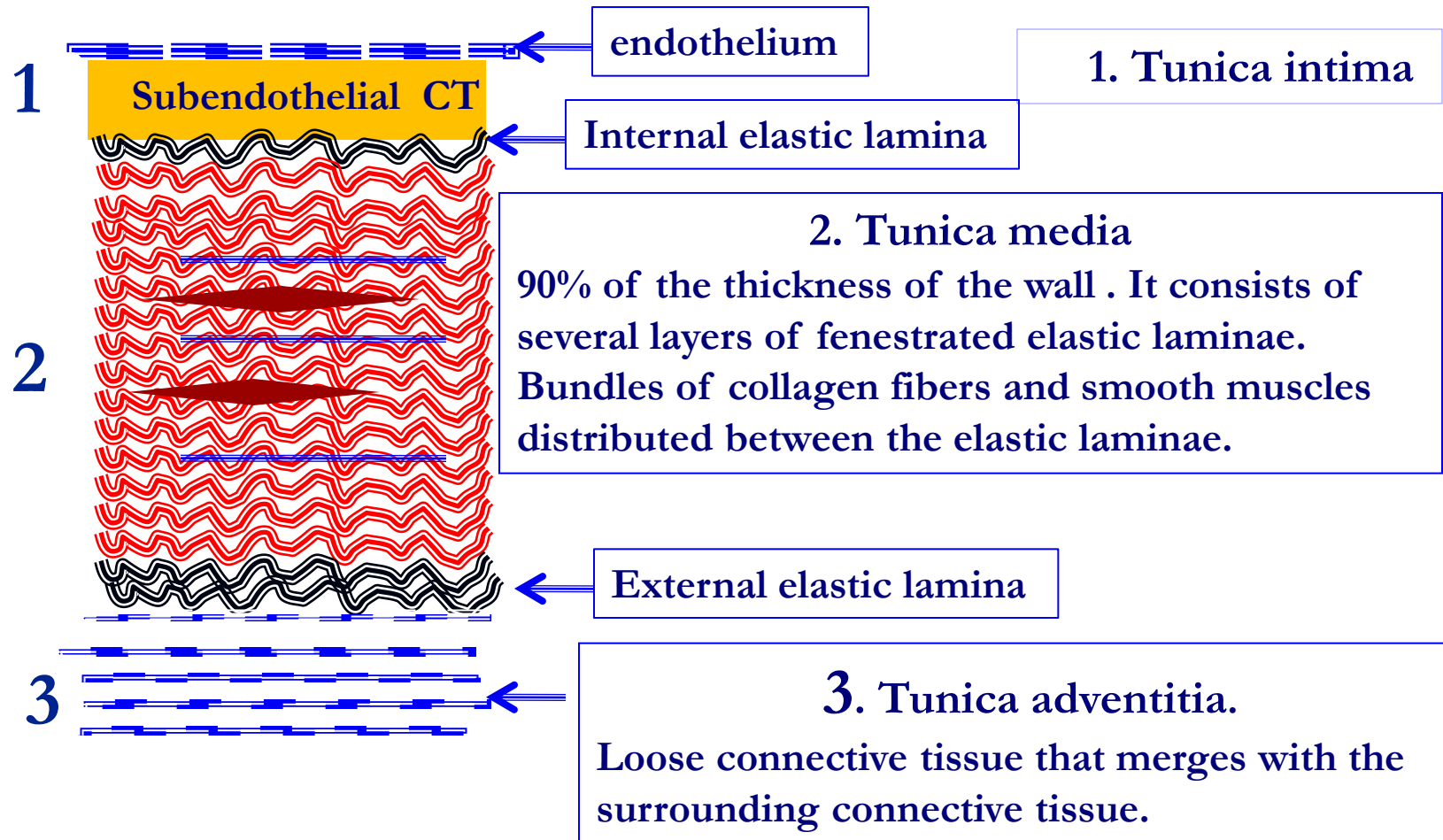
```
graph TD; A([Arteries]) --> B[Elastic<br/>(conducting)]; A --> C[Muscular<br/>(medium sized)]; A --> D[Arterioles<br/>(small sized)];
```

Elastic
(conducting)

Muscular
(medium sized)

Arterioles
(small sized)

Elastic Artery



Elastic Artery

1. Tunica intima

endothelium & subendothelial connective tissue. Internal elastic lamina is well developed, but may be **difficult to be recognize**, because it merges with the elastic fibers of the tunica media.

2. Tunica media

90% of the thickness of the wall . It consists of several layers of **fenestrated elastic laminae**. Bundles of collagen fibers and smooth muscles distributed between the elastic laminae. External elastic lamina is formed from the outer most elastic lamina, it is also **difficult to recognize**.

3. Tunica adventitia.

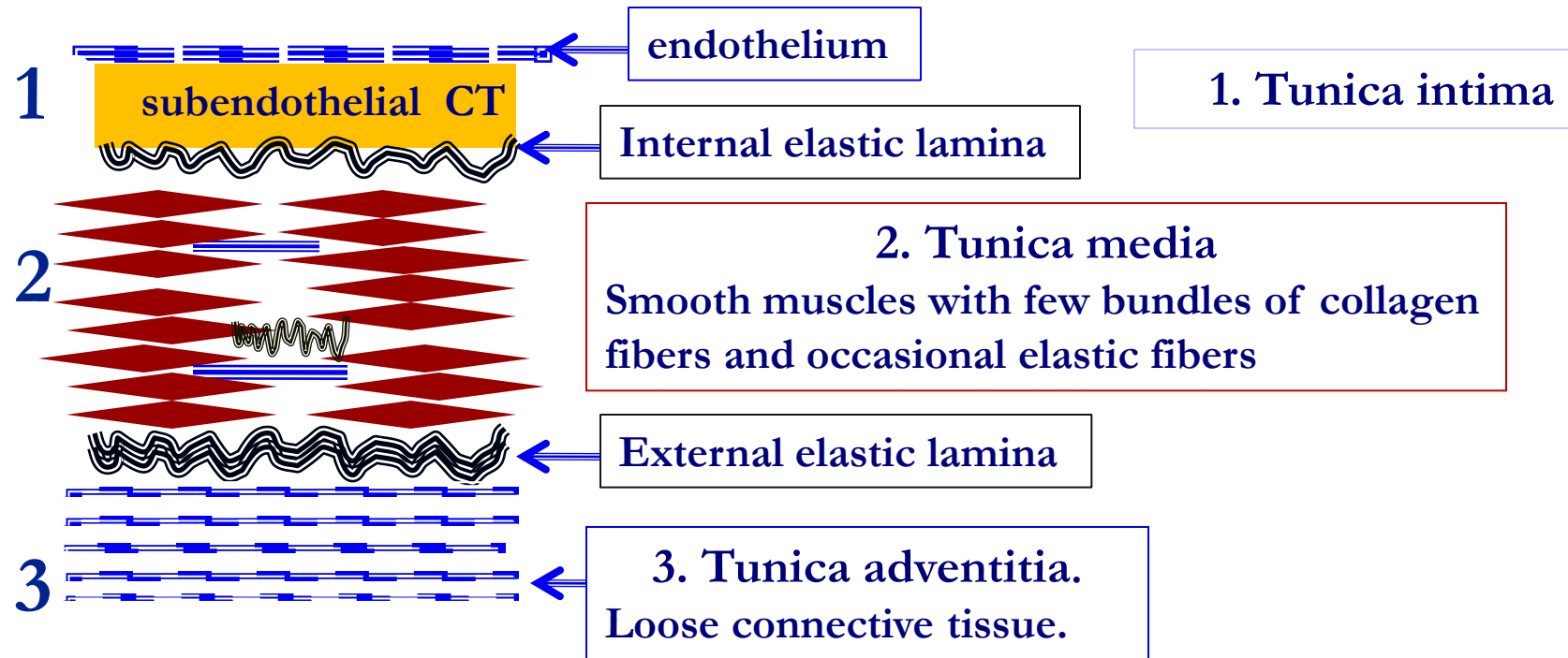
Loose connective tissue that merges with the surrounding connective tissue. It contains the vascular and nerve supply of the elastic artery.



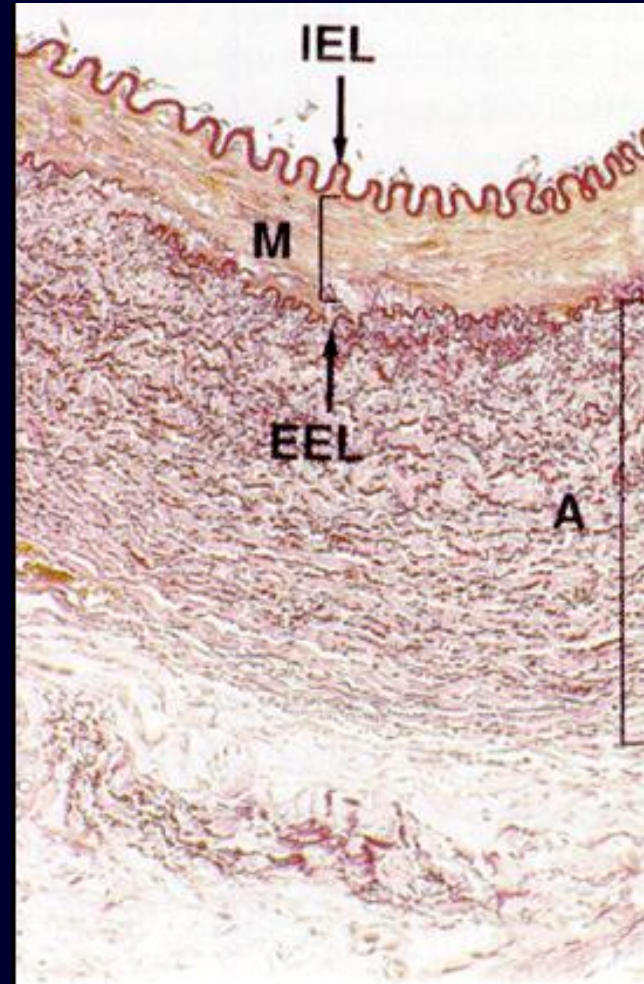
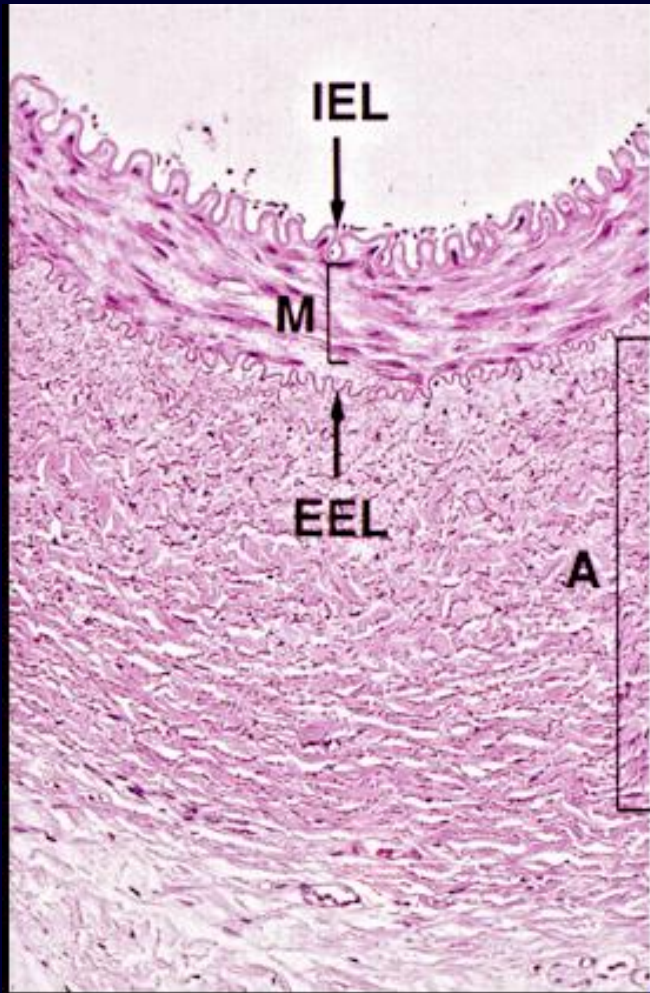
Muscular Artery

- are the medium sized arteries such as brachial & radial arteries.
- The wall of these arteries consists of:
- **Tunica intima:** endothelium rests on a subendothelial layer of connective tissue. Internal elastic lamina is well distinct.
- **Tunica media:** of concentric layers of smooth muscle cells mixed with a few bundles of collagen fibers and occasional elastic fibers. External elastic lamina is well distinct.
- **Tunica adventitia:** of loose connective tissue that merges with the surrounding connective tissues. Numerous sympathetic nerve fibers are found in the tunica adventitia. These nerve fibers stimulate the contraction of the smooth muscles in the tunica media to regulate the diameter of the lumen of the muscular arteries.

Muscular Artery



Muscular Artery



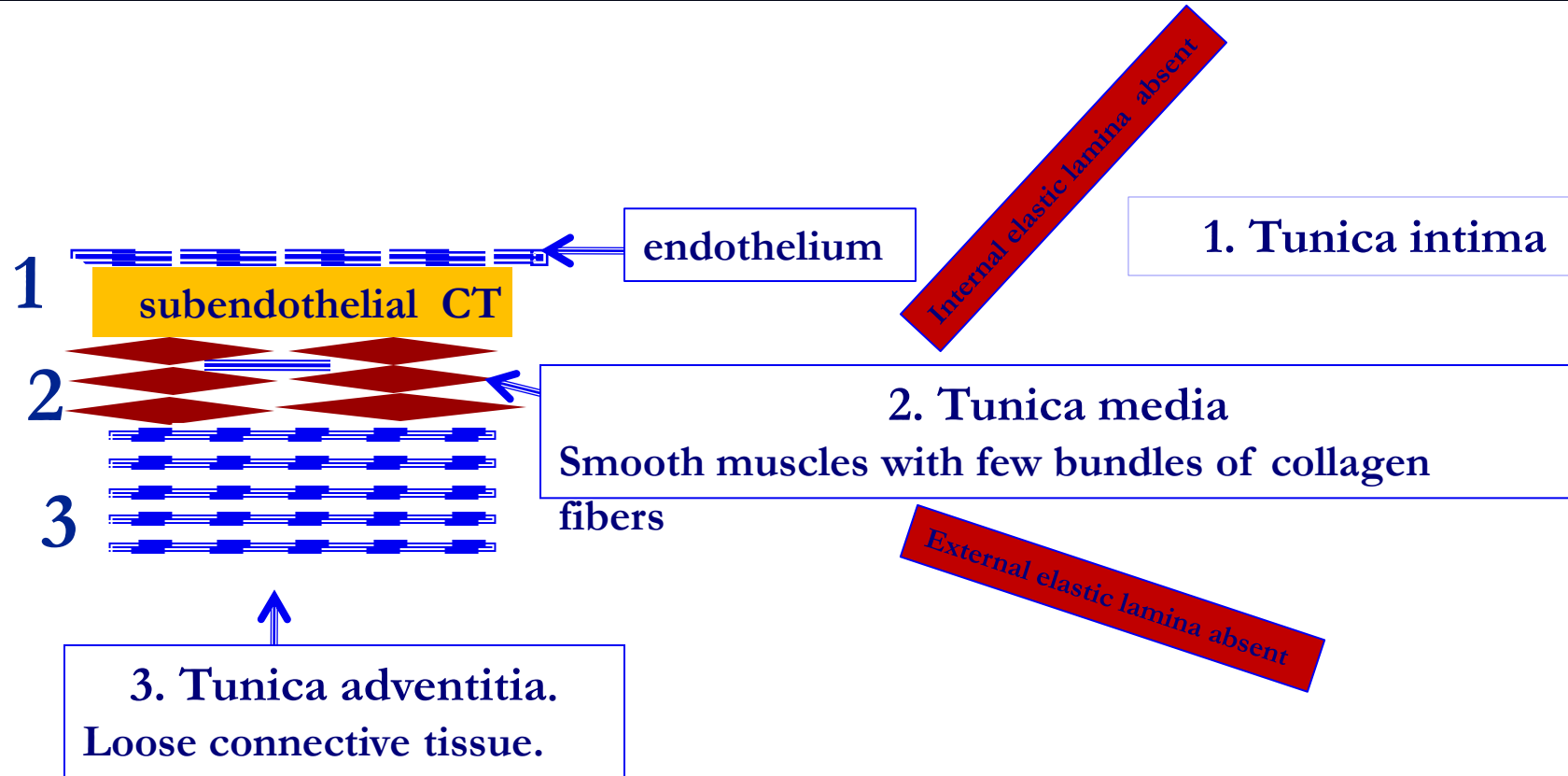
IEL = internal elastic lamina
M = Tunica media

EEL = external elastic lamina
A = tunica adventitia

Arteriole

- Is an artery having small diameter (less than 100 μm in human).
- The wall of the arterioles possesses the usual three coats.
- **Tunica intima** : of endothelium rest on a thin subendothelial connective tissue layer which is absent in small arterioles.
- Lamina elastica interna is absent.
- **Tunica media** : of circularly arranged smooth muscles. Few collagen fibers may be found.
- Lamina elastica externa is absent.
- **Tunica adventitia**: is a thin layer of loose connective tissue containing **autonomic nerve fibers** which stimulate smooth muscle contraction to control the size of the lumina of the arterioles.

Arteriole



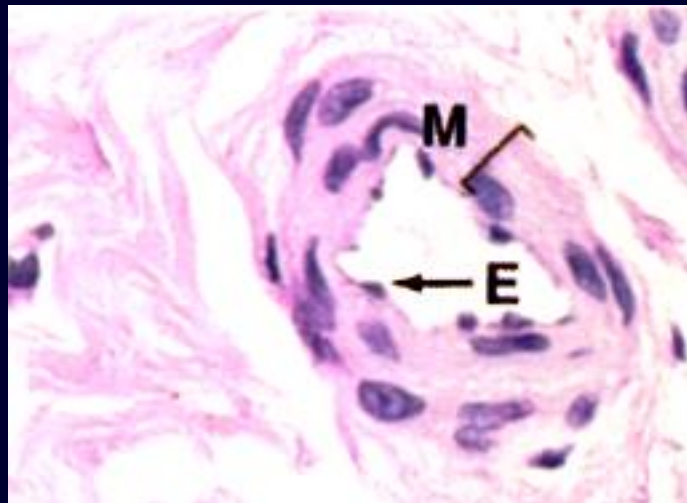
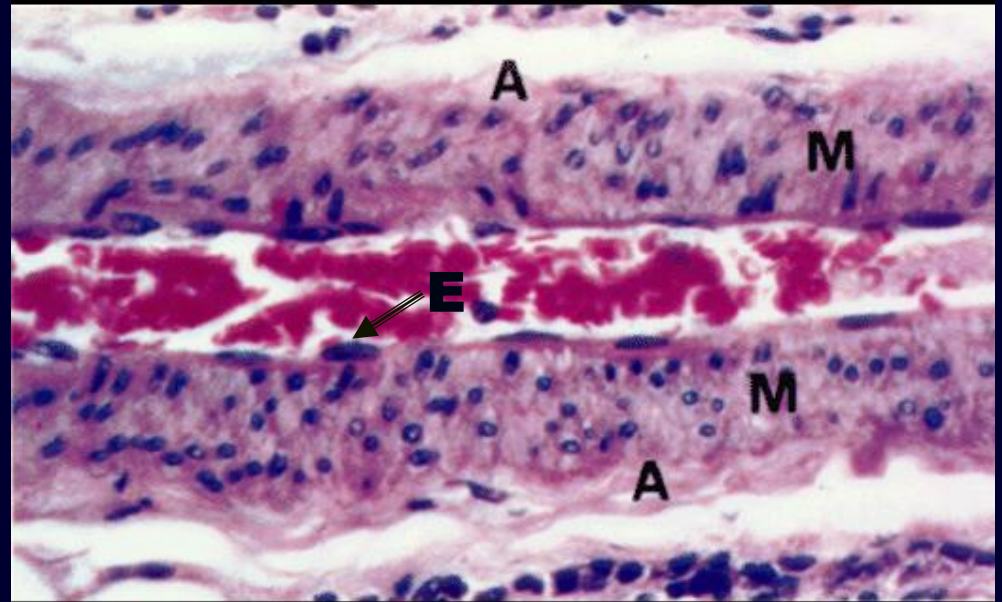
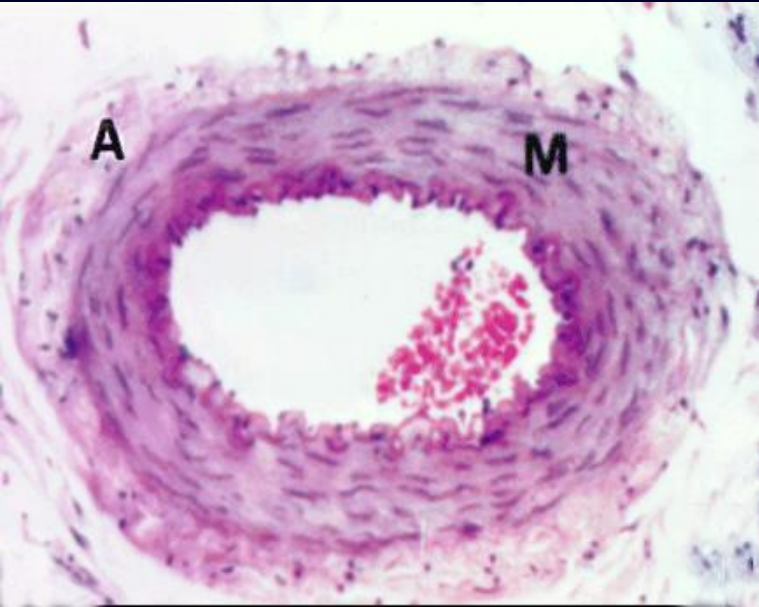
Metarterioles

- The terminal branch of many arterioles called **metarterioles** which have a small lumen and only **a thin wall lined by endothelium rest on a basal lamina.**

Tunica media composed of few smooth muscle cells.

- At the junction of metarteriole with the capillary there is a **band of smooth muscle called precapillary sphincter.**
- This sphincter regulate the blood flow to the capillaries.

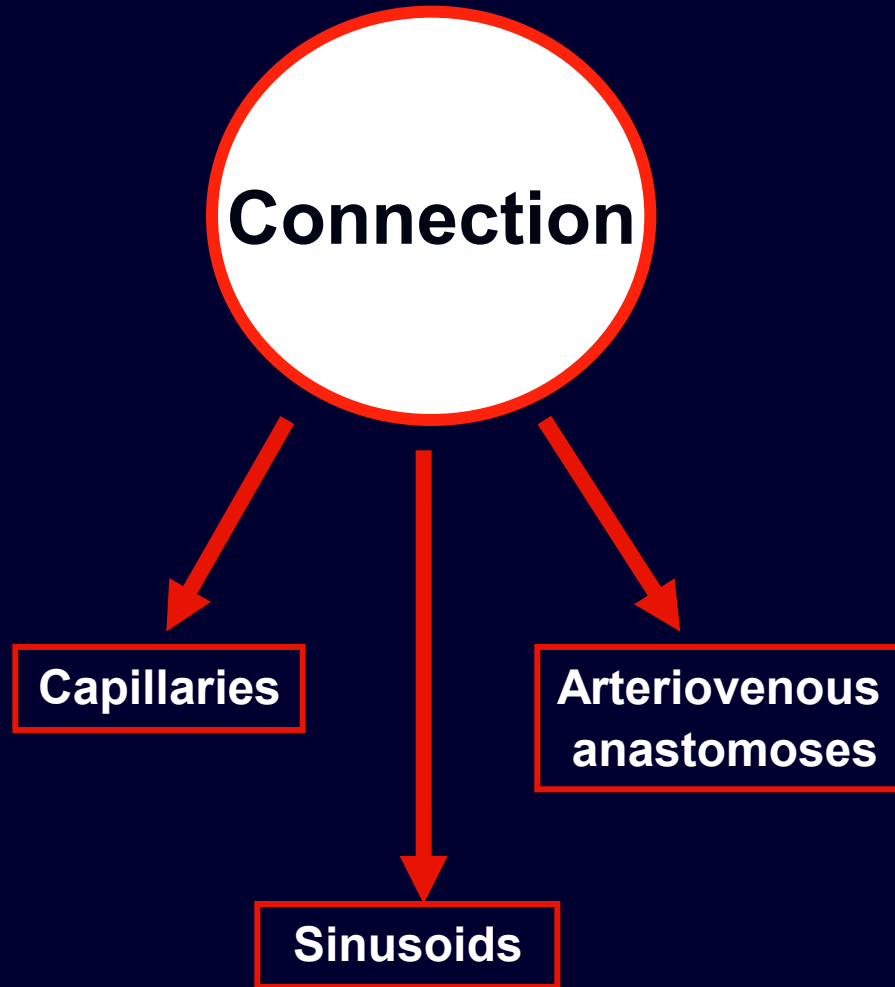
Arterioles



E = Endothelium

M = Tunica media

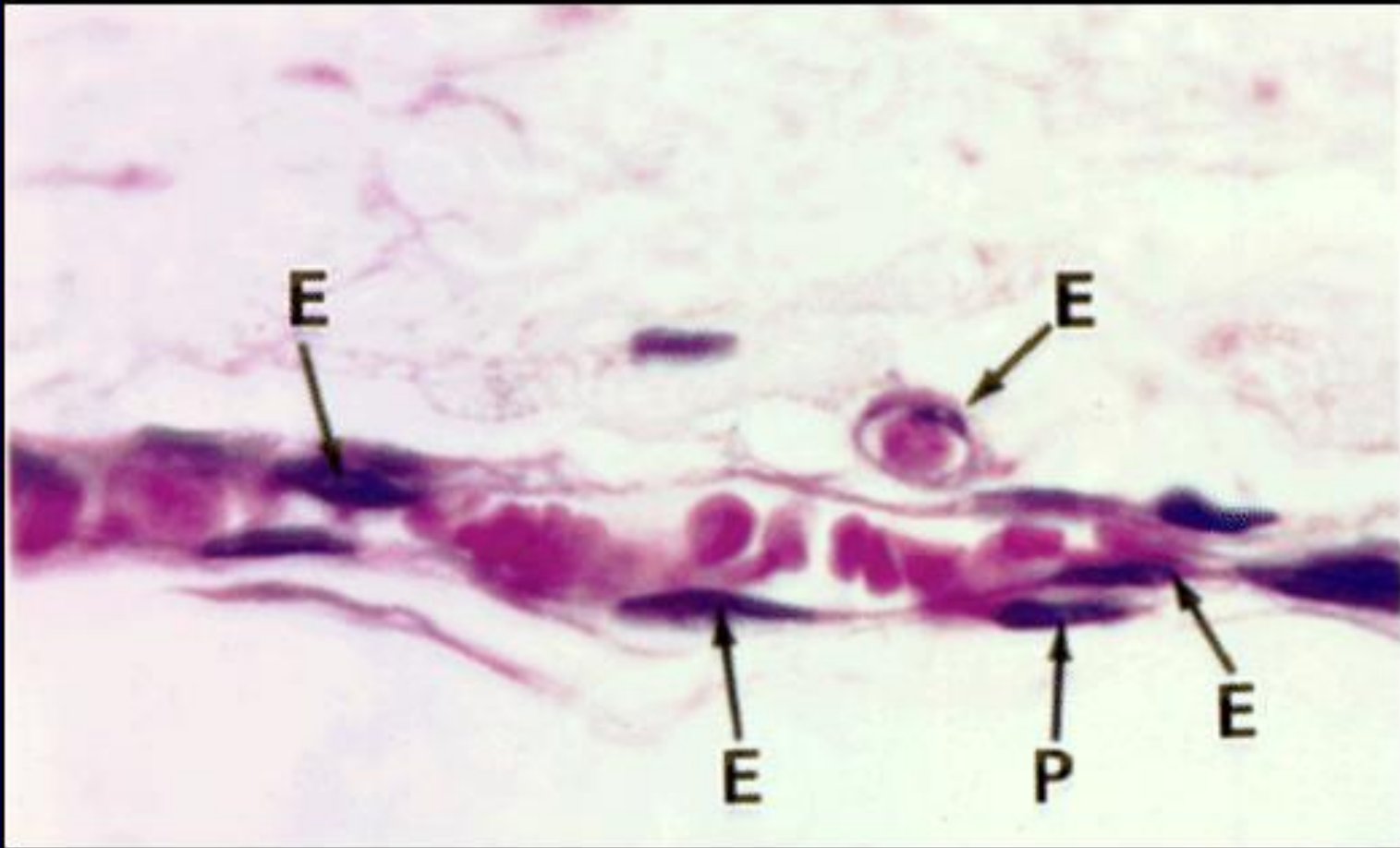
A = Tunica adventitia



Capillaries

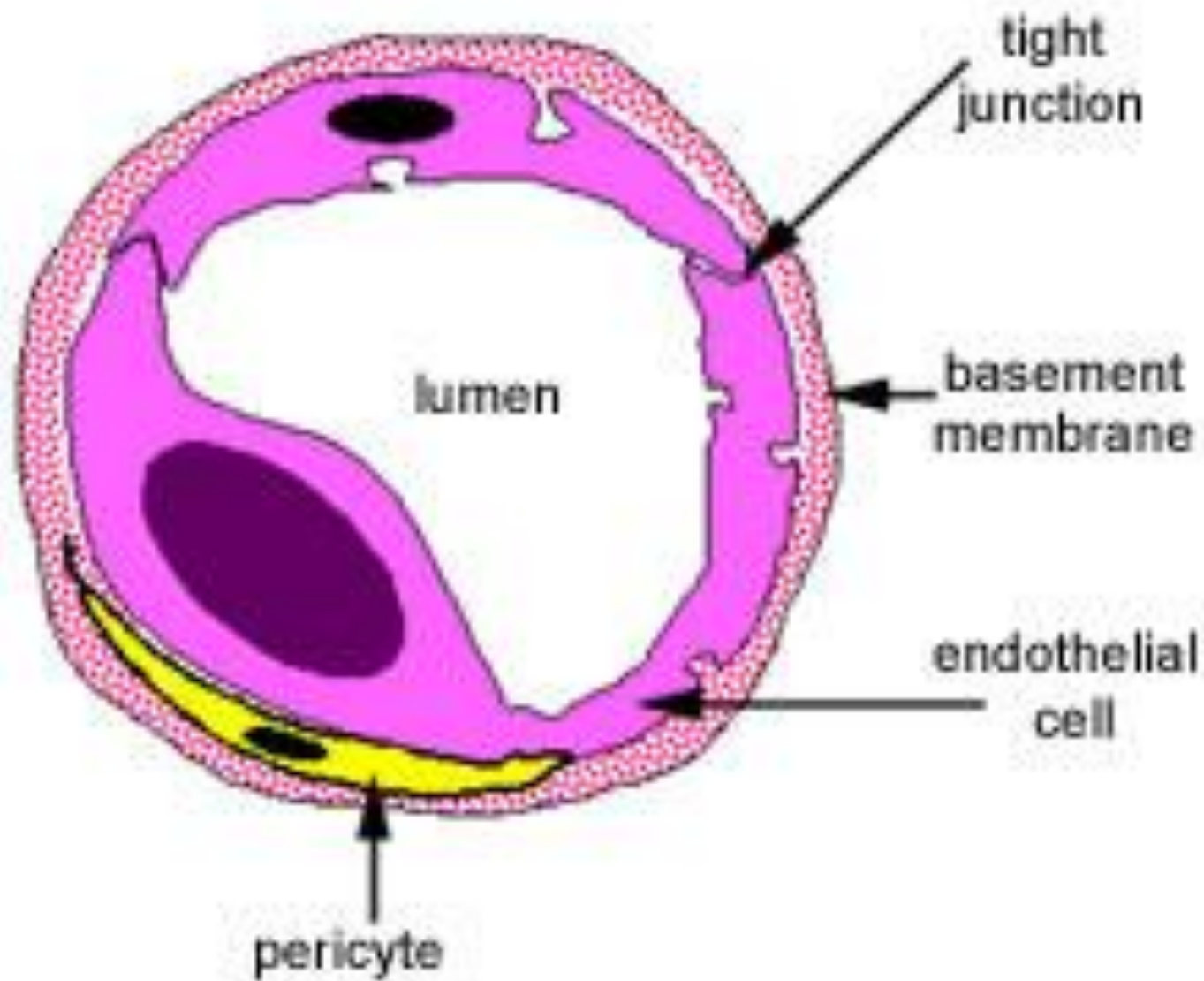
- The arterioles break up into small capillaries.
- The lumen of blood capillary is about 5 to 10 μm in diameter.
- The wall of a capillary is composed of a single layer of endothelial cells resting on a basement membrane.
- The endothelial cells are held together by tight junctions.
- **Pericytes** are undifferentiated perivascular cells surround the endothelium of blood capillary and enclosed within the basement membrane.
- **Pericytes** can differentiate into fibroblasts and smooth muscles

Capillaries



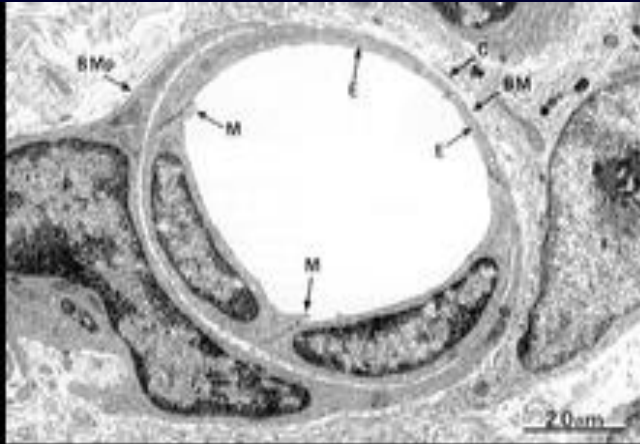
E = Endothelium

P = Pericytes



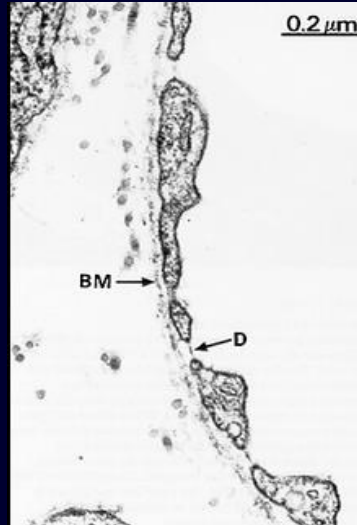
Types of Capillaries

Continuous



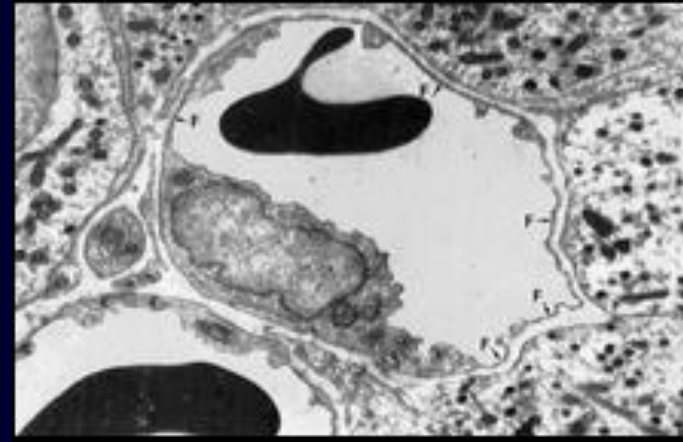
- Numerous pinocytotic vesicles:
(muscle capillaries)
- No vesicles at all:
(neural capillaries)

**Fenestrated
(visceral)**



- Circular openings or pores
(fenestrae = windows).
closed by a thin diaphragm
- Gastrointestinal tract
- Endocrine glands.

Porous

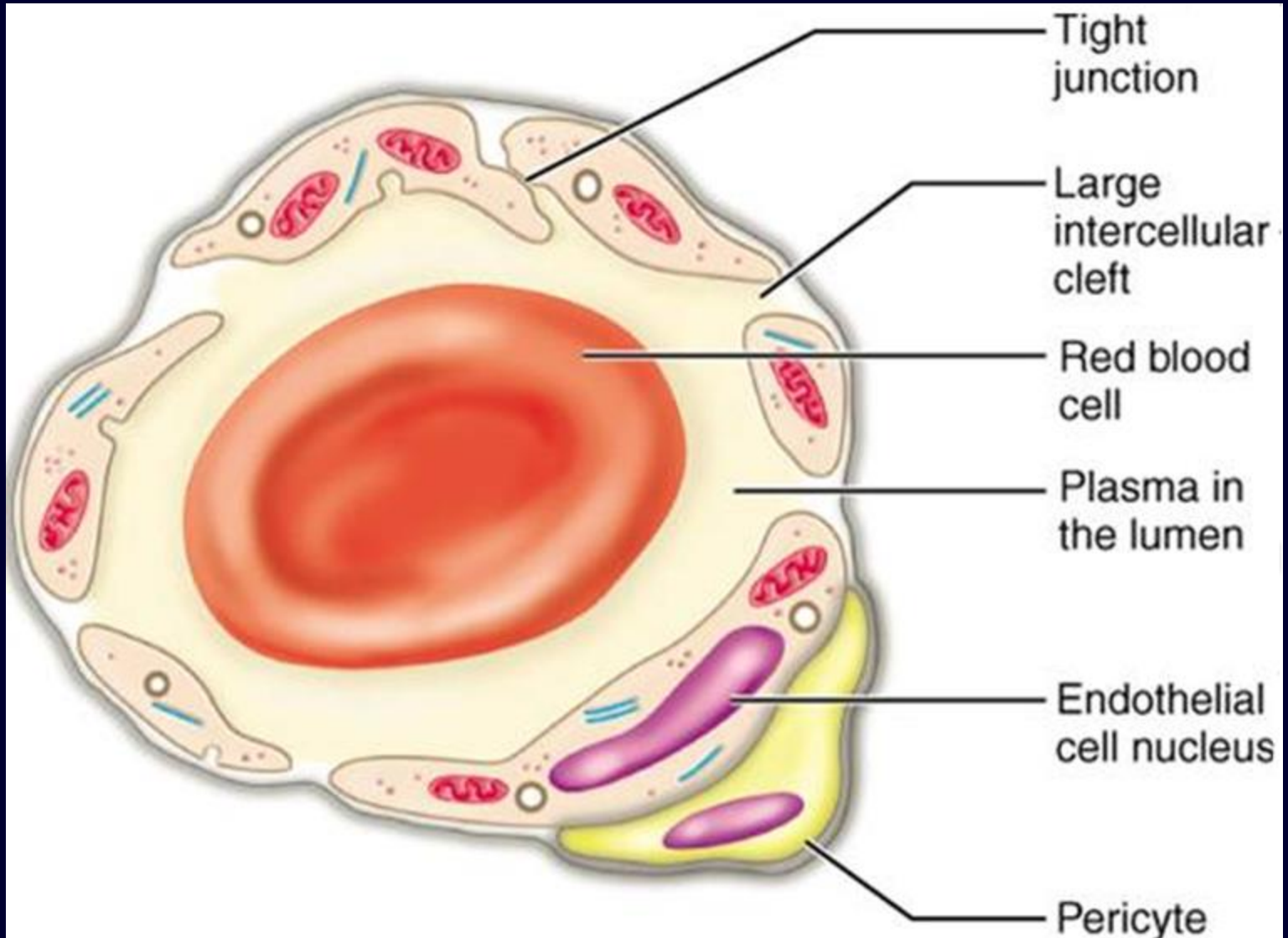


- Fenestrated but with
out diaphragm
- Kidney glomerulus.

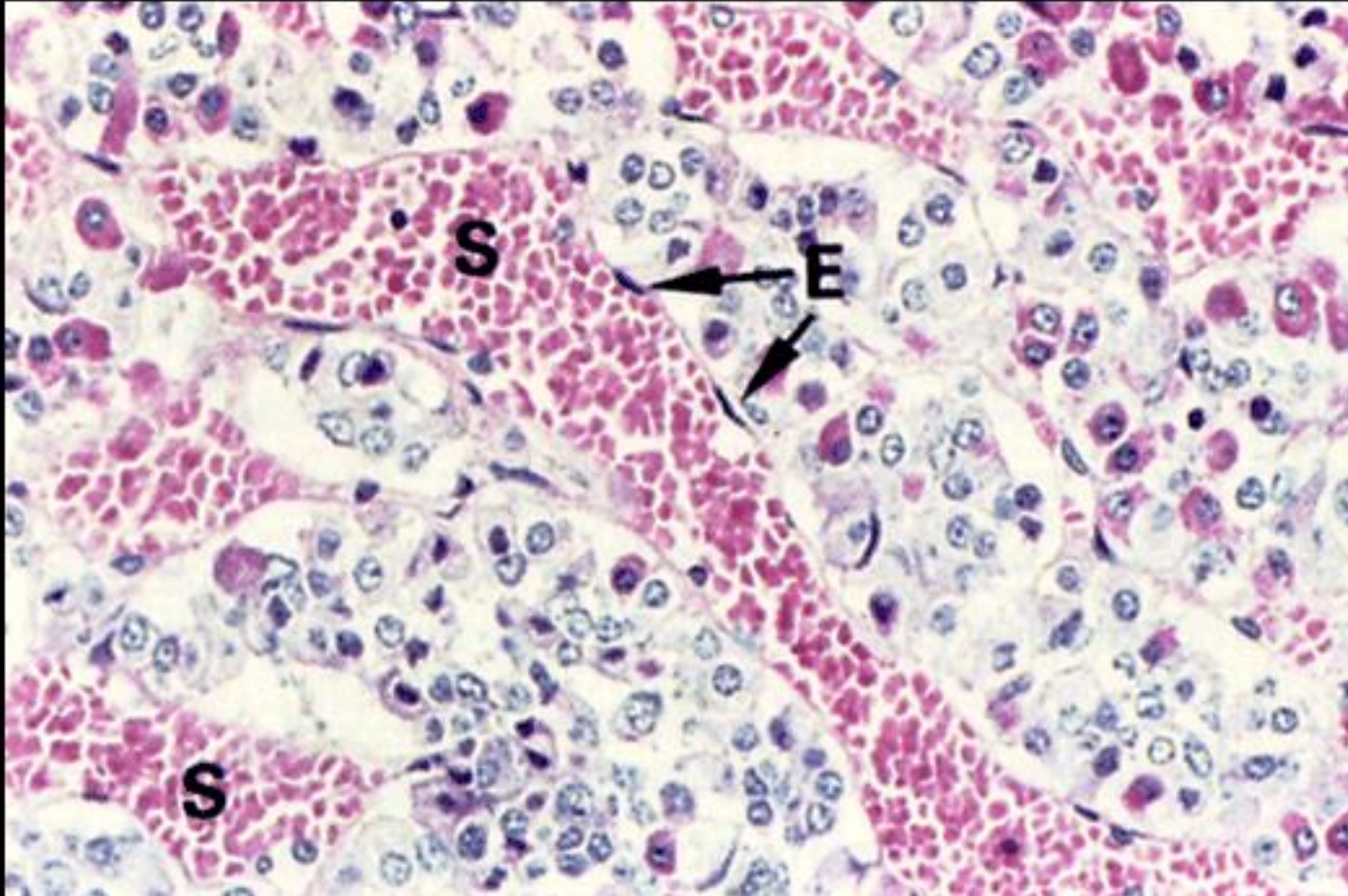
Sinusoid

- Larger than capillaries and more permeable with a thin wall and an irregular diameter.
- Lined by **endothelium resting on interrupted basement membrane** and **supported by few reticular fibers**.
- Some openings (clefts) are found between the endothelial cells where basal lamina is absent to provide a maximum exchange between blood and surrounding parenchyma and permit the passage of blood cells.
- Macrophages are found on the outer surfaces of sinusoidal walls.
- Pseudopodia of these macrophages protrude between endothelial cells into the lumen of the sinusoids.
- Macrophages in the wall of liver sinusoids called **Kupffer cells** and these of the spleen are called **Littoral cells**.

Sinusoid



Sinusoids



Arteriovenous anastomoses

- Arteriovenous anastomoses (AV shunt). means that, the arterioles are connected directly to the venules without capillaries in between. so that, the blood flows from the arterioles directly into the venules.
- The walls of the shunt are thick, muscular and well supplied with vasomotor nerve fibers.
- These muscular walls control blood flow through the venules and also through the capillaries.
- The constriction of an anastomoses wall will increase blood flow through the capillaries and vice versa.

Arteriovenous anastomoses

- The arteriovenous anastomoses are found in skin, lips, nasal mucosa, male and female reproductive system, ear and human nail beds of fingers and toes.
- Their function is to regulate the blood flow and regulate heat loss from the surrounding regions..

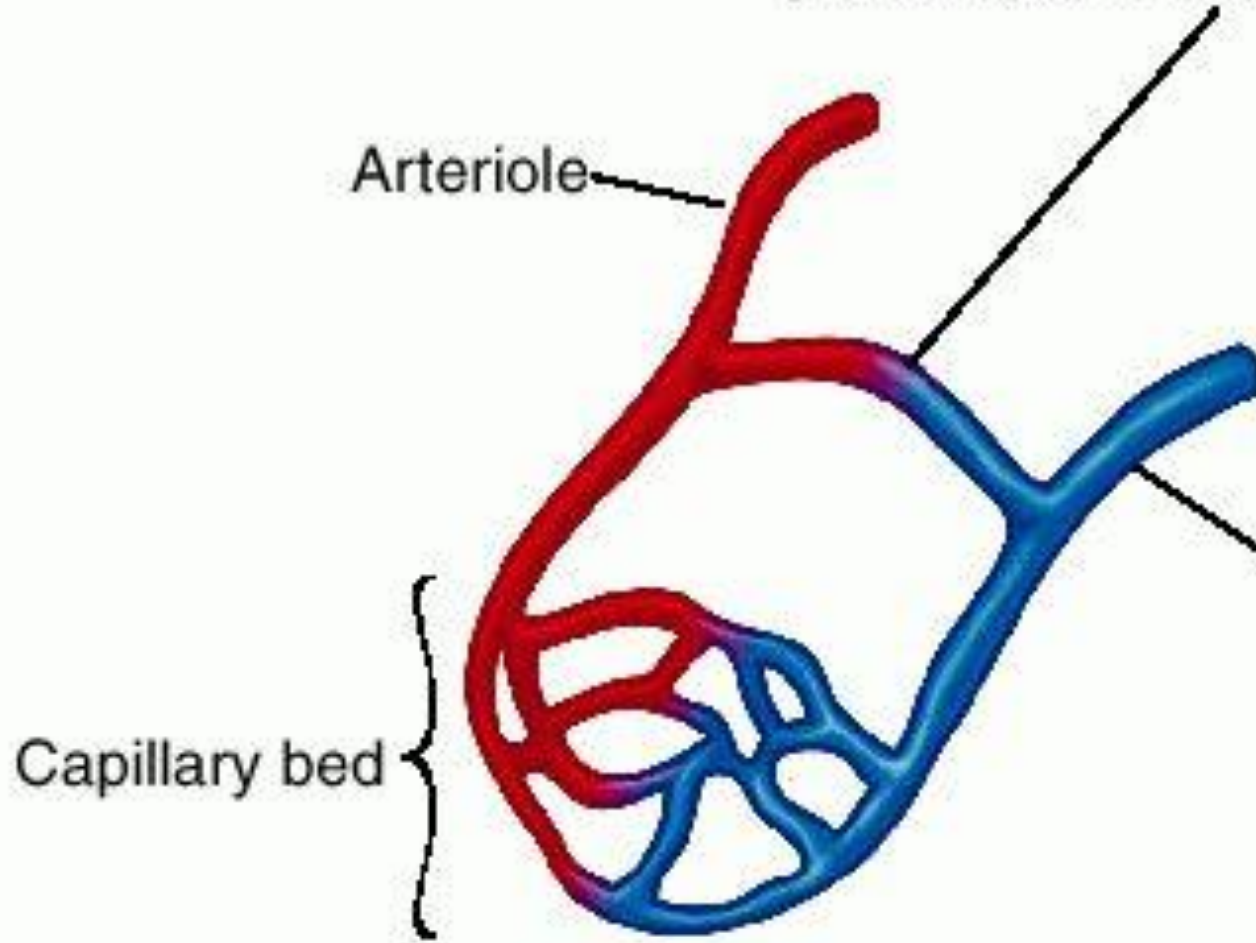
Simple arteriovenous anastomosis
(Anastomosis arteriovenosa simplex)

Arteriole

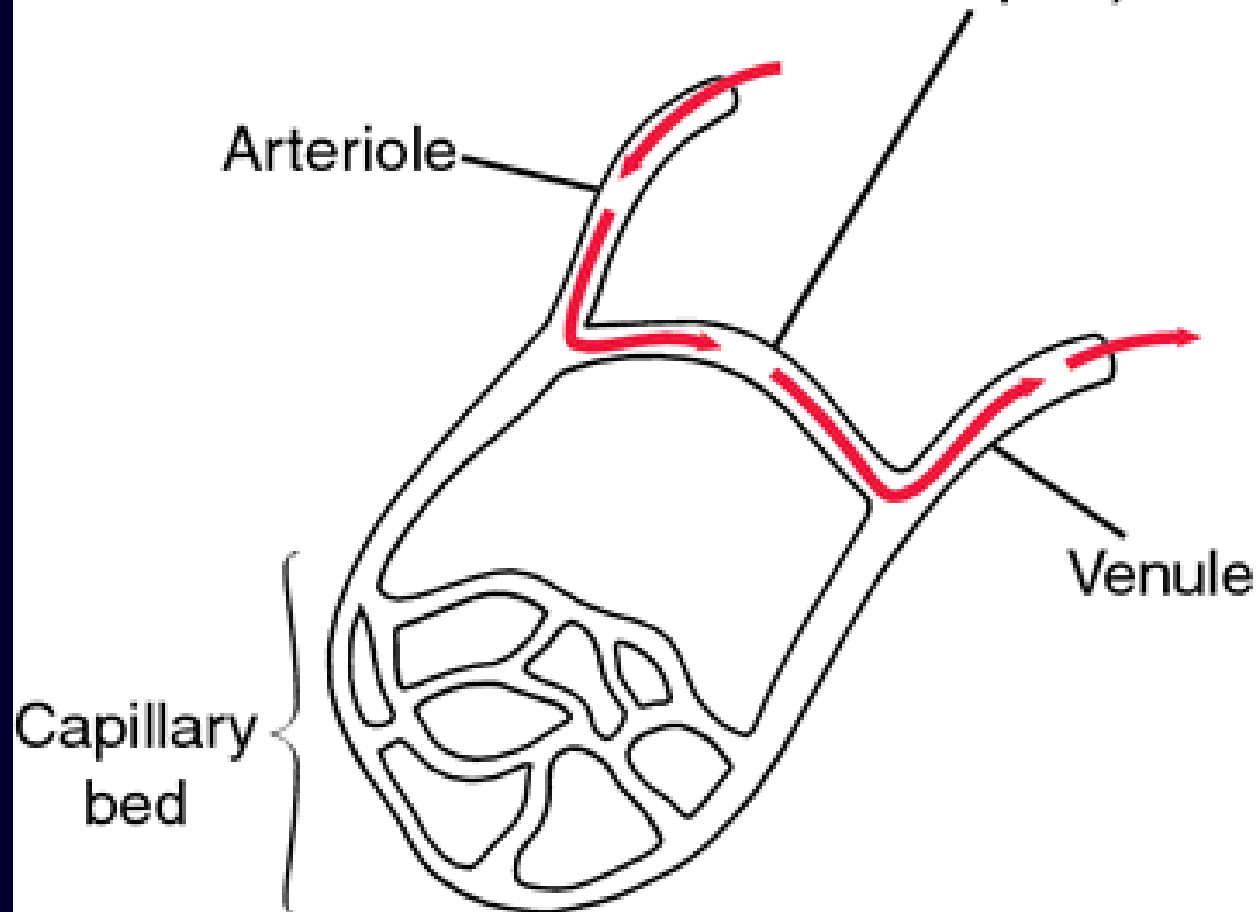
Venule

Capillary bed

■ Anastomosis arteriovenosa simplex



Simple
arteriovenous
anastomosis
(Anastomosis
arteriovenosa
simplex)



```
graph TD; Veins((Veins)) --> Venules[Venules]; Veins --> Medium["( medium sized)"]; Veins --> Large[Large]
```

Veins

Venules

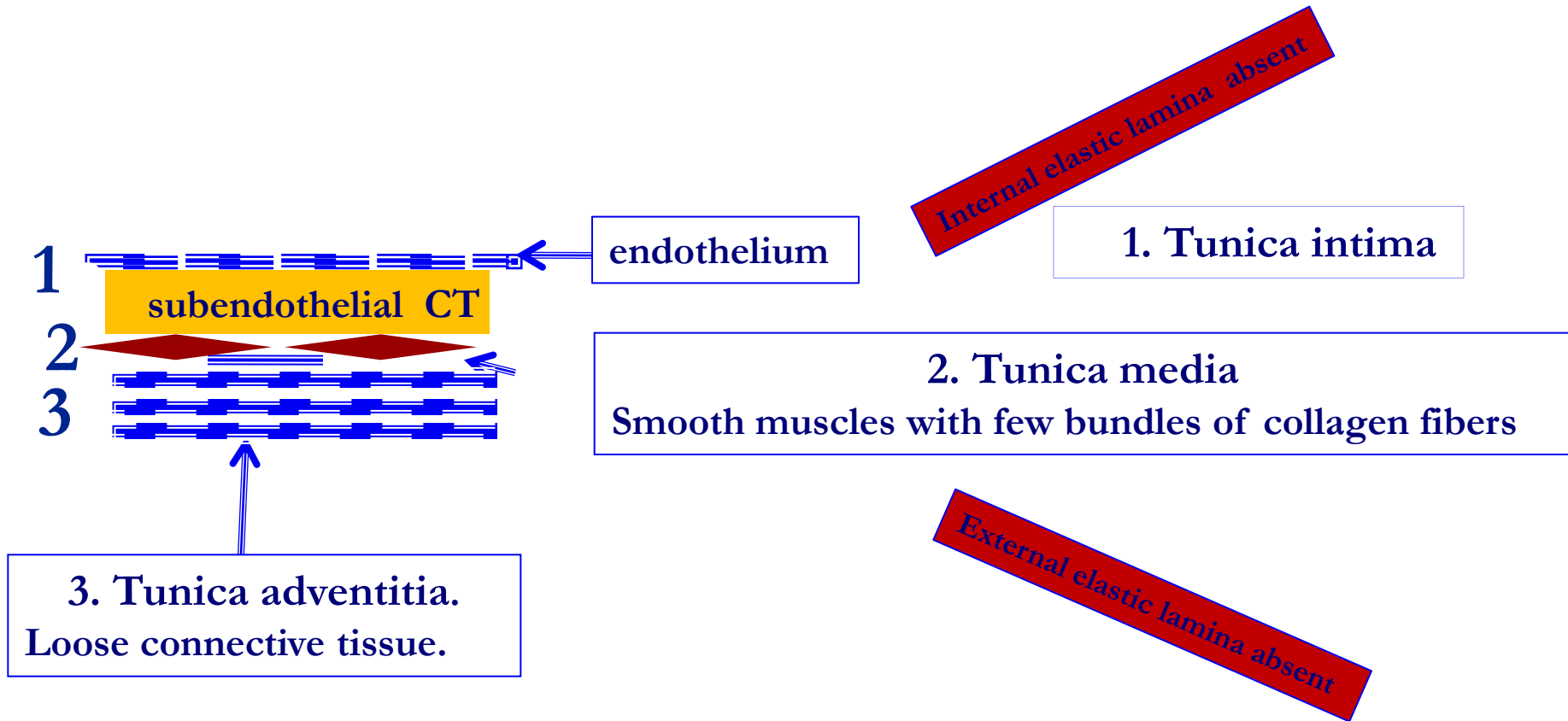
(medium sized)

Large

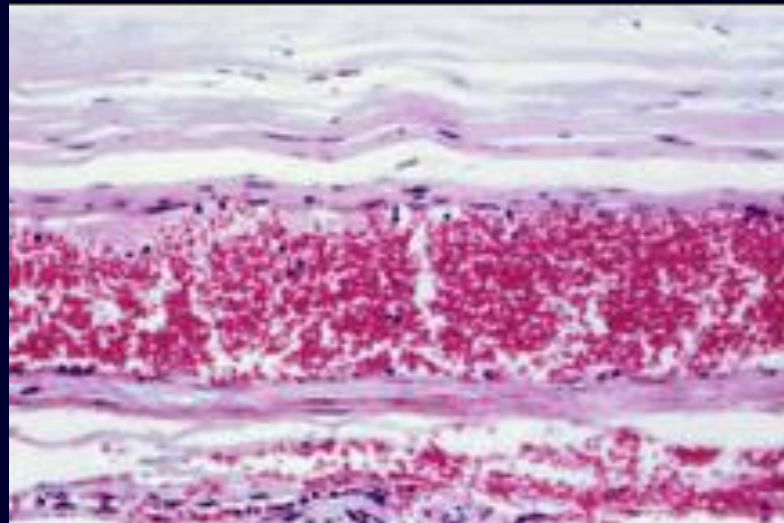
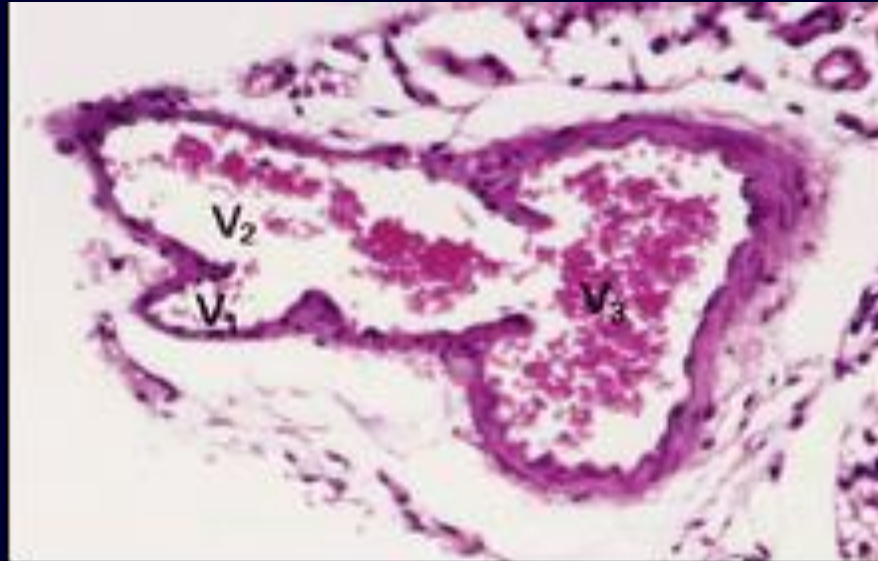
Venule

- Venules receive the blood from the capillaries and are of a larger diameter than capillaries (about 20 μm).
- The structure of a small venule is identical to that of the continuous capillary.
- Some smooth muscle fibers are situated in the connective tissue surrounding the endothelium in large venules.
- Through the wall of the capillaries and venules only, oxygen and nutritive materials pass from the blood into the tissue fluid and carbon dioxide and waste products of tissue metabolism pass into the blood.

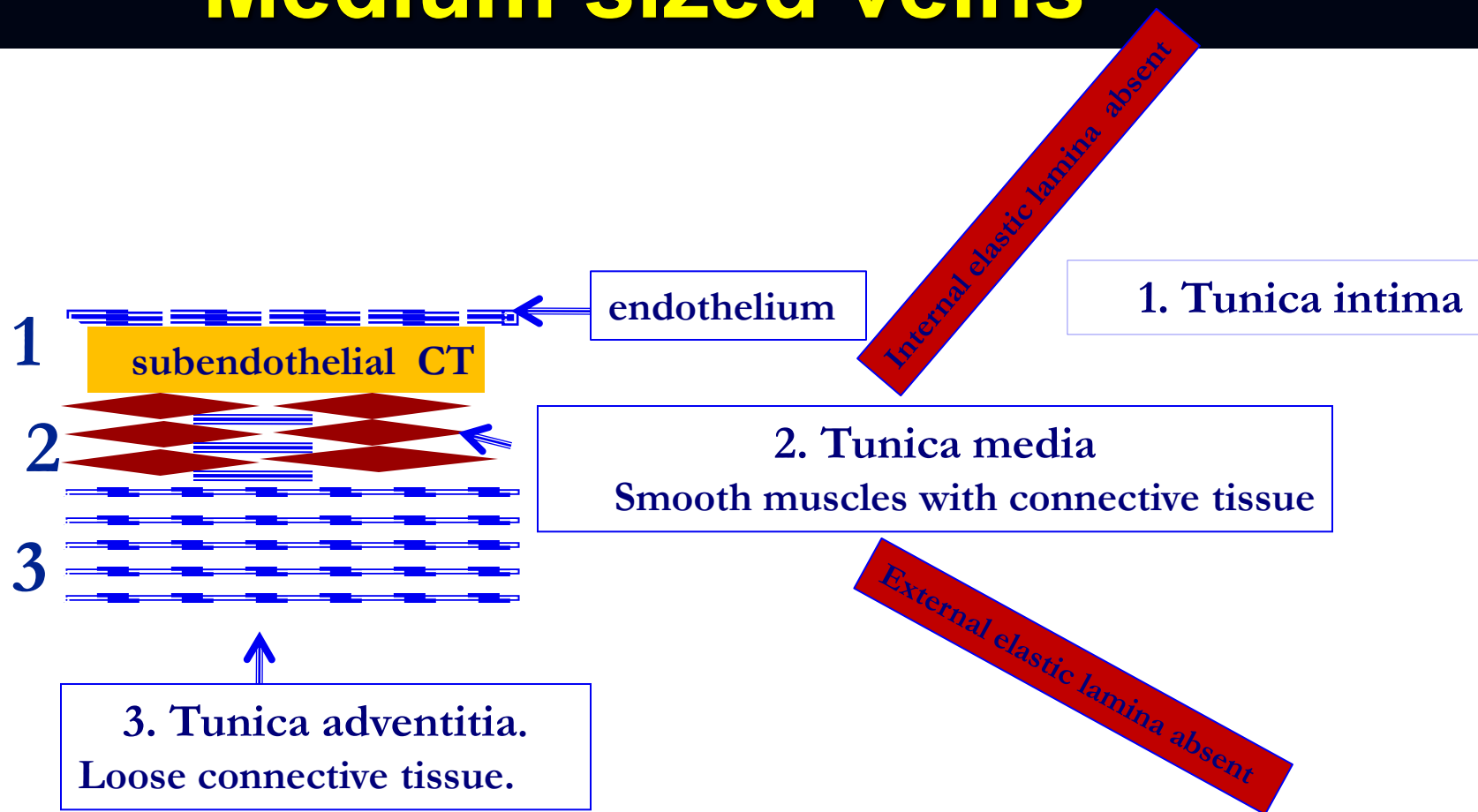
Venule



venules



Medium sized veins



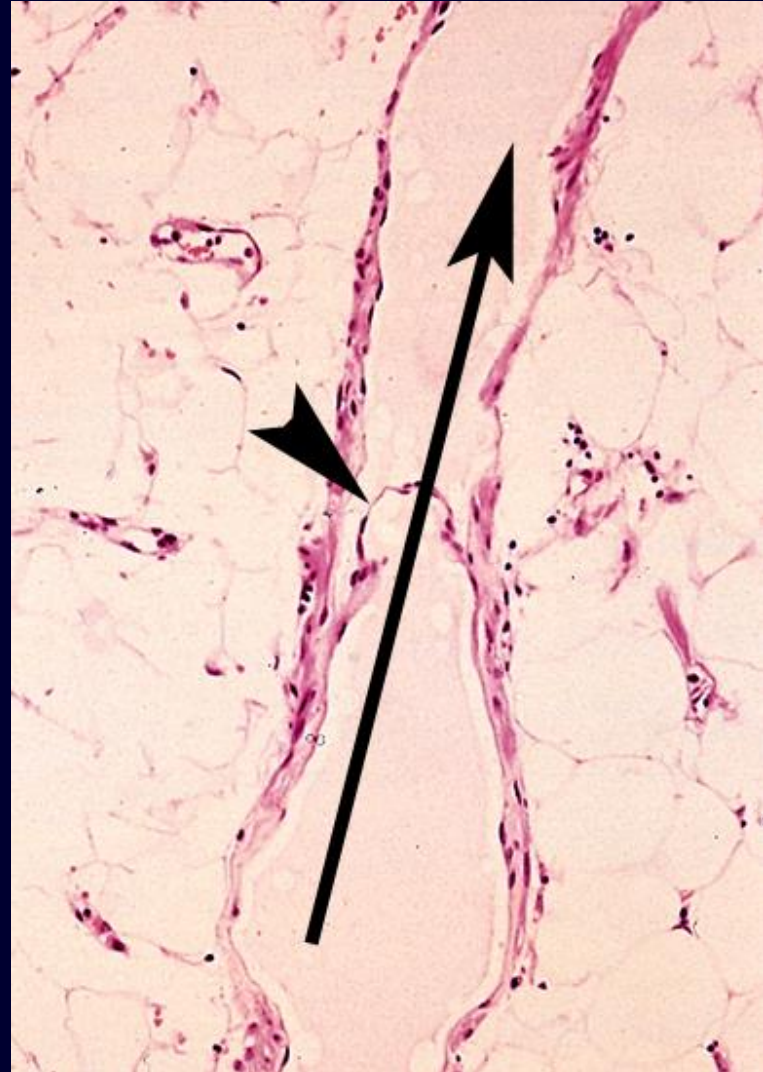
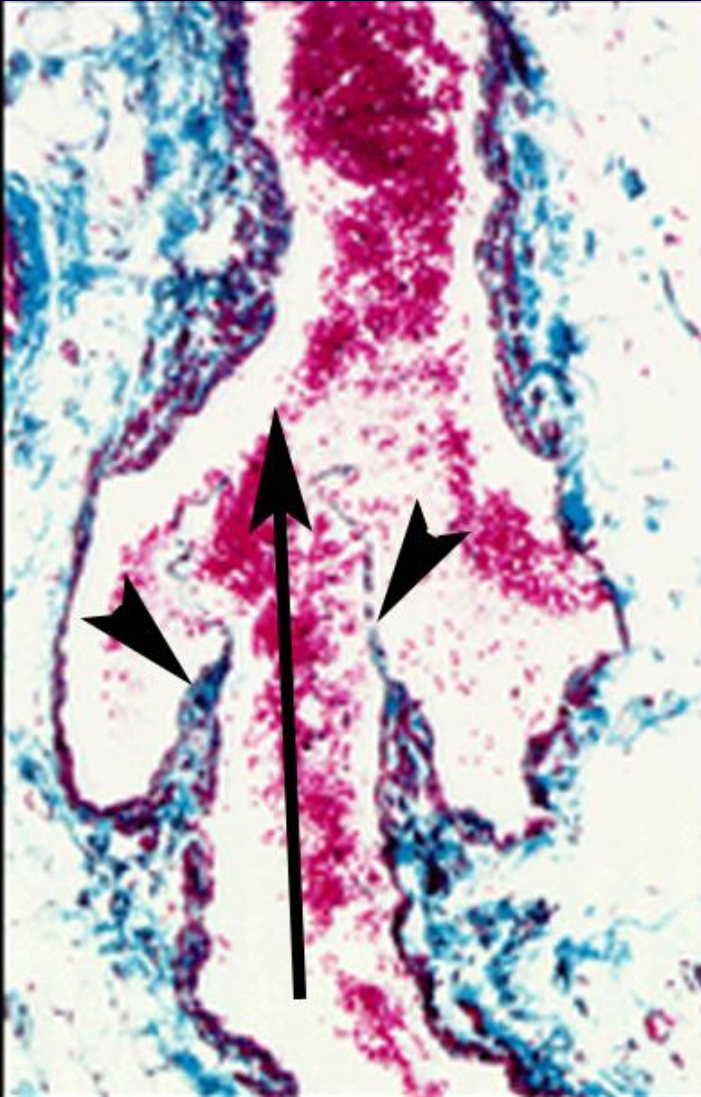
Medium sized veins

- Include the distal veins of the limbs and veins of the viscera.
- The wall consists of the three coats of blood vessels.
- **Tunica intima.** Consists of lining endothelial cells resting on a basement membrane and a thin layer of connective tissue.
- **Tunica media.** Is thinner than that found in arteries and consists of circularly arranged smooth muscle fibers mixed with connective tissue.
- **Tunica adventitia.** Is thicker than the tunica media and composed of loose connective tissue.

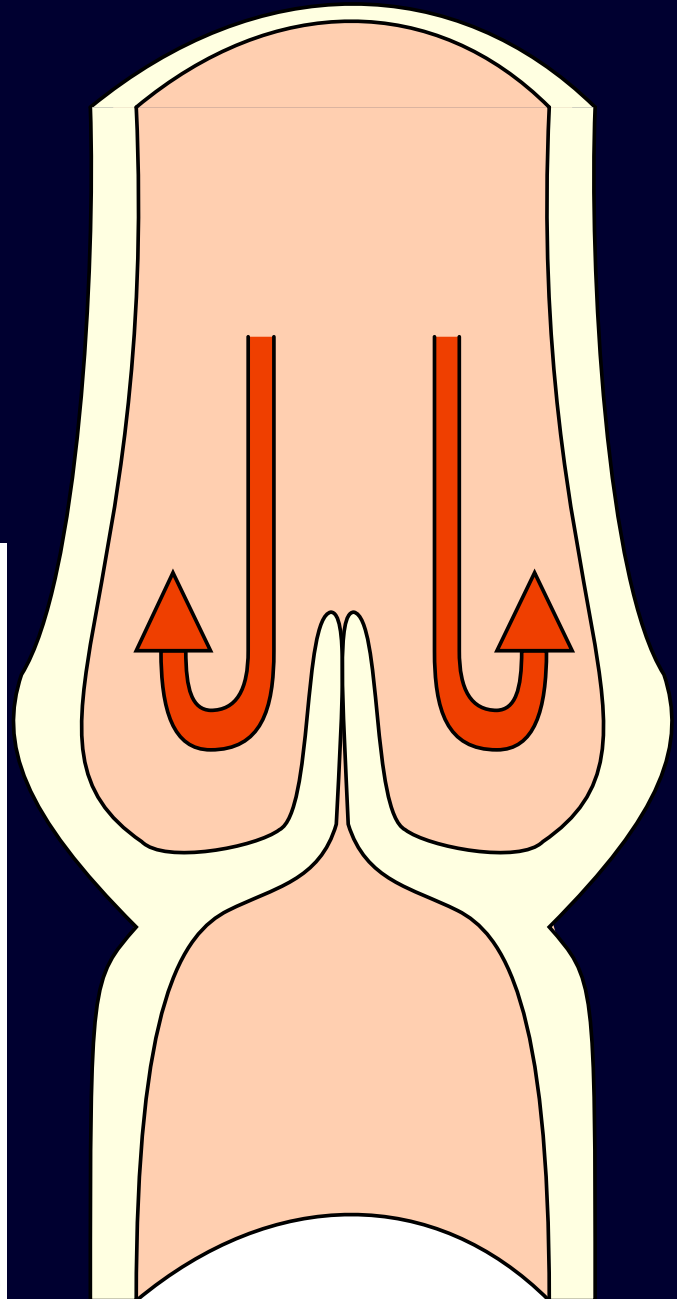
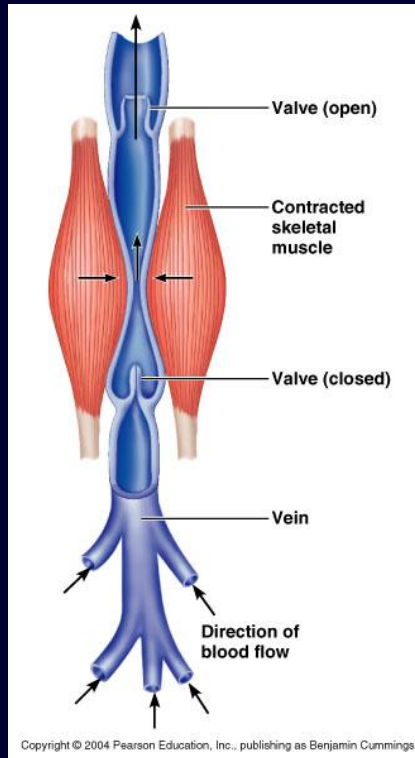
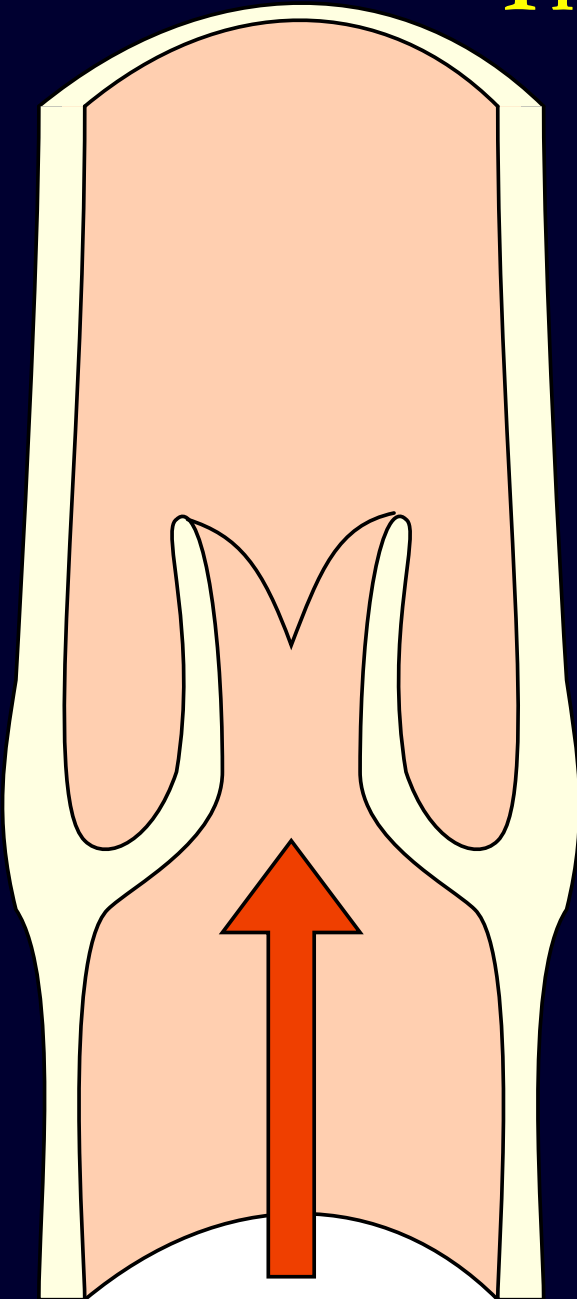
Valves of veins

- Many veins, including venules and medium sized veins of the extremities are provided with valves
- The valves allow the blood to flow toward the heart but not in the opposite direction.
- Most of the valves have two leaflets (bicuspid) and each leaflet is composed of a fold of tunica intima

Venule



The venous valves

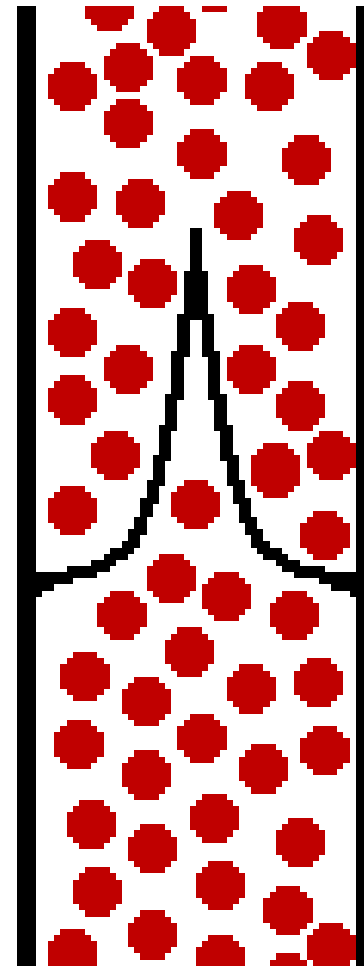




Normal Vein

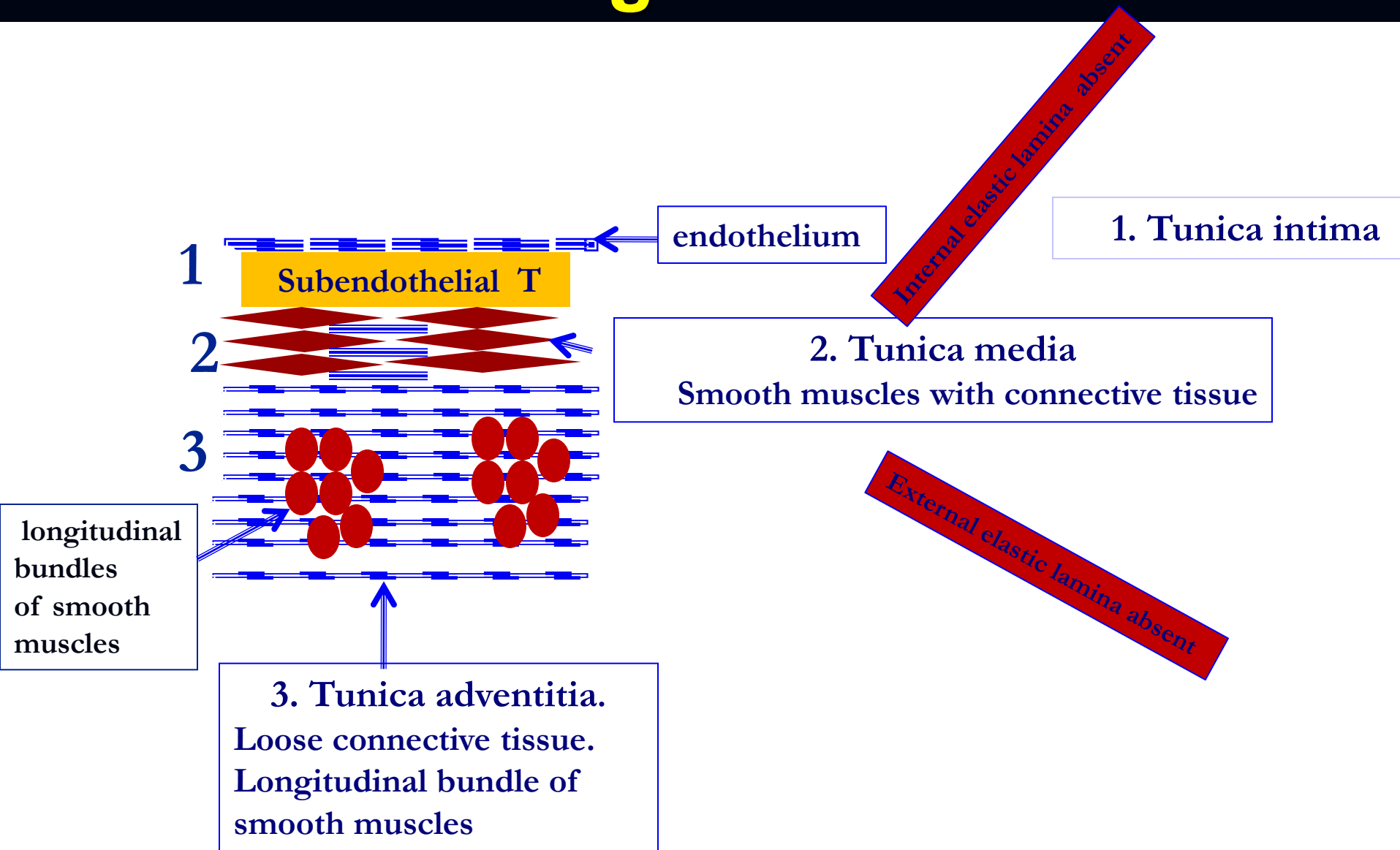


Varicose Vein



The venous valves prevent the back flow of blood.

Large veins



Large veins

- Include the **cranial and caudal venae cavae, the portal vein, and the iliac veins.**
- The Wall of these veins consists of the 3 coats:
- **Tunica intima:** of endothelial cells resting on a basement membrane and thick layer of subendothelial connective tissue.
- **Tunica media:** A thin layer of circularly arranged smooth muscle fibers mixed with connective tissue.
- **Tunica adventitia:** Is the thickest layer of the wall.
- Composed of loose connective tissue containing bundles of longitudinally arranged collagen fibers and smooth muscle cells.
- Small blood vessels, lymphatic vessels and nerves are present in this coat.

Large Vein



Tunica intima

Tunica media

Tunica adventitia

longitudinal bundles
of smooth muscles

Thank you



Dr. Mohamed Zidan
mohamedzidan@alexu.edu.eg

Good luck

كلهم لهم نفس تركيب الاتتيميا و الادفنتيشيا

الاستيك لامينا هما 2 انترنال و اكسترنال و هما فى 2 فقط الاستيك و ماسكولار
ارتري

ميديا فى كلهم سموس مصل و كلاجن و فيو سموس مصل ما عدا الاستك ارتري

الوحيد ال فية مصل و لونجتيدينال هو الارج فين